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DEVICES, SYSTEMS, AND METHODS FOR DETECTING TISSUE AND FOREIGN OBJECTS DURING A SURGICAL OPERATION

Abstract

A surgical instrument is disclosed herein. The surgical instrument can include an end effector that includes jaws configured to transition between an opened condition and a closed condition, a plurality of electrodes positioned within the jaws of the end effector, a control circuit, and a memory configured to store an algorithm configured to cause the control circuit to determine an impedance signal based on signals received from the plurality of electrodes, detect a media positioned between the jaws of the end effector based on the determined impedance signal, determine a position of the detected media based on the received signals, and generate an alert associated with the detected media and the determined position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/586,743, entitled “Devices, Systems, and Methods for Detecting Tissue and Foreign Objects During a Surgical Operation,” filed Feb. 26, 2024, which is a continuation of U.S. patent application Ser. No. 17/965,336, entitled “Devices, Systems, and Methods for Detecting Tissue and Foreign Objects During a Surgical Operation,” filed Oct. 13, 2022, issued as U.S. Pat. No. 11,957,342 on Apr. 16, 2024, which claims priority to U.S. Provisional Pat. App. No. 63/274,207, entitled “Devices, Systems, and Methods for Detecting Tissue and Foreign Objects During a Surgical Operation,” filed Nov. 1, 2021, and to U.S. Provisional Pat. App. No. 63/330,502, entitled “Devices, Systems, and Methods for Detecting Tissue and Foreign Objects During a Surgical Operation,” filed Apr. 13, 2022, the disclosure of each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present invention relates to surgical instruments and, in various arrangements, to surgical stapling and cutting instruments and staple cartridges for use therewith that are designed to staple and cut tissue.

SUMMARY

[0003] In one aspect, a surgical instrument is disclosed. The surgical instrument can include an end effector that includes jaws configured to transition between an opened condition and a closed condition, a plurality of electrodes positioned within the jaws of the end effector, a control circuit, and a memory configured to store an algorithm configured to cause the control circuit to determine an impedance signal based on signals received from the plurality of electrodes, detect a media positioned between the jaws of the end effector based on the determined impedance signal, determine a position of the detected media based on the received signals, and generate an alert associated with the detected media and the determined position.

[0004] In one aspect, a surgical system is disclosed. The surgical system can include a surgical instrument including an end effector including jaws configured to transition between an opened condition and a closed condition, and a plurality of electrodes positioned within the jaws of the end effector, wherein each electrode of the plurality of electrodes is positioned about a longitudinal axis defined by the end effector, and a computer system communicably coupled to the surgical instrument, wherein the computer system includes a control circuit and a memory configured to store an algorithm configured to cause the control circuit to receive signals from the plurality of electrodes, determine an impedance signal based on the signals received from the plurality of

electrodes, detect a media positioned between the jaws of the end effector based on the determined impedance signal, determine a position of the detected media along the longitudinal axis based on the signals received from the plurality of electrodes, and generate an alert associated with the detected media and the determined position.

[0005] In one aspect, a method of characterizing media positioned between jaws of an end effector of a surgical instrument is disclosed. The method can include receiving, via a control circuit of the surgical instrument, signals from a plurality of electrodes positioned within the jaws of the end effector, determining, via the control circuit, an impedance signal based on the signals received from the plurality of electrodes, detecting, via the control circuit, the media positioned between the jaws of the end effector based on the determined impedance signal, determining, via the control circuit, a position of the detected media along a longitudinal axis defined by the end effector based on the signals received from the plurality of electrodes, generating, via the control circuit, an alert associated with the detected media and the determined position, and characterizing, via the control circuit, the detected media based on the determined impedance signal.

[0006] In addition to the foregoing, various other method and/or system and/or program product aspects are set forth and described in the teachings such as text (e.g., claims and/or detailed description) and/or drawings of the present disclosure.

[0007] The foregoing is a summary and thus may contain simplifications, generalizations, inclusions, and/or omissions of detail; consequently, those skilled in the art will appreciate that the summary is illustrative only and is not intended to be in any way limiting. Other aspects, features, and advantages of the devices and/or processes and/or other subject matter described herein will become apparent in the teachings set forth herein.

[0008] In one or more various aspects, related systems include but are not limited to circuitry and/or programming for effecting herein-referenced method aspects; the circuitry and/or programming can be virtually any combination of hardware, software, and/or firmware configured to affect the herein-referenced method aspects depending upon the design choices of the system designer. In addition to the foregoing, various other method and/or system aspects are set forth and described in the teachings such as text (e.g., claims and/or detailed description) and/or drawings of the present disclosure.

[0009] Further, it is understood that any one or more of the following-described forms, expressions of forms, examples, can be combined with any one or more of the other following-described forms, expressions of forms, and examples.

[0010] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

Description

FIGURES

[0011] The novel features of the described forms are set forth with particularity in the appended claims. The described forms, however, both as to organization and methods of operation, may be best understood by reference to the following description, taken in conjunction with the accompanying drawings in which:

[0012] FIG. 1 illustrates a perspective view of a surgical instrument in accordance with at least one embodiment;

[0013] FIG. 2 illustrates a perspective view of a controller of a robotic surgical system;

[0014] FIG. 3 illustrates a perspective view of the robotic surgical system of FIG. 2 comprising a plurality of robotic surgical arms which each operably support a surgical instrument thereon;

[0015] FIG. 4 illustrates a side view of a robotic surgical arm illustrated in FIG. 3;

[0016] FIG. 5 illustrates a perspective view of a staple cartridge positioned in a cartridge jaw in accordance with at least one embodiment;

[0017] FIG. 5A illustrates a partial cross-sectional view of the staple cartridge of FIG. 5;

[0018] FIG. 5B illustrates a perspective view of the staple cartridge of FIG. 5 removed from the cartridge jaw;

[0019] FIG. 5C illustrates an exploded view of the staple cartridge of FIG. 5;

[0020] FIG. 5D illustrates a perspective view of a sled of the staple cartridge of FIG. 5;

[0021] FIG. 6 illustrates a logic flow diagram of an algorithm depicting a control program or a logic configuration for modulating a sensor parameter of the sensor array, in accordance with at least one aspect of the present disclosure;

[0022] FIG. 7 illustrates a top schematic view of a staple cartridge, in accordance with at least one aspect of the present disclosure;

[0023] FIG. 8 illustrates a diagram of a cartridge comprising a plurality of sensors coupled to a control circuit through a set of coils to transfer power and data between the cartridge and a control circuit located in an instrument housing, in accordance with at least one aspect of the present disclosure;

[0024] FIG. 9 illustrates a block diagram of a surgical instrument configured or programmed to control the distal translation of a displacement member, in accordance with at least one aspect of the present disclosure;

[0025] FIG. 10 illustrates a system for detecting tissue and foreign objects during a surgical operation, in accordance with at least one non-limiting aspect of the present disclosure;

[0026] FIG. 11 illustrates a user interface displayed by a computing device of the system of FIG. 10, in accordance with at least one non-limiting aspect of the present disclosure;

[0027] FIG. 12 illustrates a chart depicting a parameter measured by the system of FIG. 10 in time, in accordance with at least one non-limiting aspect of the present disclosure;

[0028] FIG. 13 illustrates a logic flow diagram of a method of detecting and locating media within jaws of an end effector of a surgical instrument of the system of FIG. 10, in accordance with at least one non-limiting aspect of the present disclosure;

[0029] FIG. 14 illustrates a logic flow diagram of a method of detecting an anomaly as part of the method of FIG. 13, in accordance with at least one non-limiting aspect of the present disclosure;

[0030] FIG. 15 illustrates a logic flow diagram of a method of detecting a presence of a foreign object within jaws of an end effector, as part of the methods of FIGS. 13 and 14, in accordance with at least one non-limiting aspect of the present disclosure;

[0031] FIG. 16 illustrates a chart depicting experimental results of the methods of FIGS. 13-15, in accordance with at least one non-limiting aspect of the present disclosure;

[0032] FIGS. 17A-C illustrate another chart depicting experimental results of the methods of FIGS. 13-15, in accordance with at least one non-limiting aspect of the present disclosure;

[0033] FIG. 18 illustrates another user interface displayed by a computing device of the system of FIG. 10, in accordance with at least one non-limiting aspect of the present disclosure;

[0034] FIG. 19 illustrates another chart depicting experimental results of the methods of FIGS. 13-15, in accordance with at least one non-limiting aspect of the present disclosure;

[0035] FIG. 20 illustrates another chart depicting experimental results of the methods of FIGS. 13-15, in accordance with at least one non-limiting aspect of the present disclosure;

[0036] FIG. 21 illustrates a logic flow diagram of a method that collectively implements multiple patterns to distinguish detected foreign objects from detected tissue within the jaws of an end effector, in accordance with at least one non-limiting aspect;

[0037] FIG. 22 illustrates a logic flow diagram of a method of training an algorithmic model to intelligently classify the location of media within the jaws of an end effector, in accordance with at least one non-limiting aspect;

[0038] FIG. 23 illustrates a block diagram of a method of classifying a detected media after an algorithmic model has been trained via the method of FIG. 22, in accordance with at least one non-limiting aspect of the present disclosure;

[0039] FIGS. 24A and 24B illustrate several charts depicting a distinction between various media detected within the jaws of an end effector, as determined via the method of FIG. 23, in accordance with at least one non-limiting aspect of the present disclosure;

[0040] FIG. 25 illustrates another chart depicting a distinction between various media detected within the jaws of an end effector, as determined via the method of FIG. 23, in accordance with at least one non-limiting aspect of the present disclosure;

[0041] FIGS. 26A and 26B illustrate several charts depicting a of characterizing tissue within the jaws of an end effector, in accordance with at least one non-limiting aspect of the present disclosure;

[0042] FIG. 27 illustrates a flow chart of a method of detecting media, locating media, and characterizing media positioned between the jaws of an end effector, in accordance with at least one non-limiting aspect of the present disclosure;

[0043] FIG. 28 illustrates a surgical system configured to detect, locate, and characterize media positioned between the jaws of an end effector, in accordance with at least one non-limiting aspect of the present disclosure;

[0044] FIGS. 29A-C illustrates a robotic surgical instrument configured for use with the system of FIG. 28, in accordance with at least one aspect of the present disclosure;

[0045] FIG. 30 illustrates a block diagram of an algorithmic engine employed by the surgical instrument of FIGS. 29A-C, in accordance with at least one non-limiting aspect of the present disclosure;

[0046] FIG. 31 illustrates a handheld surgical instrument configured for use with the system of FIG. 28, in accordance with at least one non-limiting aspect of the present disclosure;

[0047] FIG. 32 illustrates an exemplary electrode array, in accordance with at least one non-limiting aspect of the present disclosure;

[0048] FIGS. 33A-C illustrates another end effector configured for use with the surgical instruments disclosed herein, in accordance with at least one non-limiting aspect of the present disclosure;

[0049] FIGS. 34A-C illustrate an articulation joint, shaft, nozzle, and control circuit configured for use with the end effectors and surgical instruments disclosed herein, according to at least one non-limiting aspect of the present disclosure;

[0050] FIG. 35 illustrates another end effector, in accordance with at least one non-limiting aspect of the present disclosure;

[0051] FIGS. 36A-D illustrate another end effector, in accordance with at least one non-limiting aspect of the present disclosure;

[0052] FIGS. 37A and 37B illustrate other end effectors, in accordance with at least one non-limiting aspect of the present disclosure;

[0053] FIG. 38 illustrates another end effector, in accordance with at least one non-limiting aspect of the present disclosure;

[0054] FIGS. 39A-D illustrate another end effector, in accordance with at least one non-limiting aspect of the present disclosure;

[0055] FIG. 40 illustrates a flexible circuit configured for use with any of the end effectors disclosed herein, in accordance with at least one non-limiting aspect of the present disclosure;

[0056] FIG. 41 illustrates a system diagram of a system configured to use the surgical instruments and end effectors disclosed herein, in accordance with at least one non-limiting aspect of the present disclosure;

[0057] FIG. 42 illustrates a surgical instrument, in accordance with at least one non-limiting aspect of the present disclosure;

[0058] FIG. 43 illustrates a diagram illustrating several non-limiting surgical system configurations, in accordance with at least one non-limiting aspect of the present disclosure; [0059] FIG. 44 illustrates a logic flow diagram of a method of identifying a particular contact between the jaws of an end effector and a media positioned within the jaws of the end effector, in accordance with at least one non-limiting aspect of the present disclosure; [0060] FIG. 45 illustrates a logic flow diagram of a method of utilizing the sensing techniques disclosed herein to inform an operation performed by a second surgical instrument, in accordance with at least one non-limiting aspect of the present disclosure; and [0061] FIG. 46 illustrates a logic flow diagram of a specific procedure implementing the method of FIG. 45, in accordance with at least one non-limiting aspect of the present disclosure. [0062] Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate various embodiments of the invention, in one form, and such exemplifications are not to be construed as limiting the scope of the invention in any manner.

DESCRIPTION

[0063] Before explaining various forms of surgical devices in detail, it should be noted that the illustrative forms are not limited in application or use to the details of construction and arrangement of parts illustrated in the accompanying drawings and description. The illustrative forms may be implemented or incorporated in other forms, variations and modifications, and may be practiced or carried out in various ways. Further, unless otherwise indicated, the terms and expressions utilized herein have been chosen for the purpose of describing the illustrative forms for the convenience of the reader and are not for the purpose of limitation thereof. As used herein, the term “surgical device” is used interchangeably with the term “surgical instrument.”

[0064] Further, it is understood that any one or more of the following-described forms, expressions of forms, examples, can be combined with any one or more of the other following-described forms, expressions of forms, and examples.

[0065] Various forms are directed to improved ultrasonic and/or electrosurgical (RF) instruments configured for effecting tissue treating, dissecting, cutting, and/or coagulation during surgical procedures. In one form, a combined ultrasonic and electrosurgical device may be configured for use in open surgical procedures, but has applications in other types of surgery, such as minimally invasive laparoscopic, orthoscopic, or thoracoscopic procedures, for example, non-invasive endoscopic procedures, either in hand held or and robotic-assisted procedures. Versatility is achieved by selective application of multiple energy modalities simultaneously, independently, sequentially, or combinations thereof. For example, versatility may be achieved by selective use of ultrasonic and electrosurgical energy (e.g., monopolar or bipolar RF energy) either simultaneously, independently, sequentially, or combinations thereof.

[0066] Although handheld and robotic surgical devices, such as surgical staplers, can provide numerous surgical benefits, it would be beneficial if such devices could be outfitted with sensing and feedback features, which could generate information regarding tissue location in jaws, tissue characteristics, and the presence of foreign objects within the jaws of a surgical device. Such features could bring high value to both hand-held and robotic surgical devices that could make surgical operations, such as stapling tasks, more efficient, precise, and safer for the patient. Electrical impedance spectroscopy (“EIS”) is a powerful technique that utilizes particularly configured signals to probe the impedance characteristics of objects. Accordingly, EIS techniques can be implemented to scan a tissue sample within the jaws of a surgical device using signals with a wide range of frequencies to generate an impedance spectrum for the tissue sample. Accordingly, there is a need for improved surgical devices, systems, and methods for detecting tissue locations and foreign objects. Such surgical devices, systems, and methods can employ EIS techniques to scan and characterize a tissue sample and thus, improve the efficiency, precision, and safety of a surgical operation.

[0067] A surgical instrument **10000** is illustrated in FIG. 1. The surgical instrument **10000** comprises a handle **10100** including a handle housing **10120**, a shaft **10200** extending from the handle **10100**, and an end effector **10400**. The end effector **10400** comprises a first jaw **10410** configured to receive a staple cartridge and a second jaw **10420** movable relative to the first jaw **10410**. The second jaw **10420** comprises an anvil including staple forming pockets defined therein. The surgical instrument **10000** further comprises a closure actuator **10140** configured to drive a closure system of the surgical instrument **10000** and move the second jaw **10420** between an unclamped position and a clamped position. The closure actuator **10140** is operably coupled with a closure tube **10240** that is advanced distally when the closure actuator **10140** is closed. In such instances, the closure tube **10240** contacts the second jaw and cams and/or pushes the second jaw **10420** downwardly into its clamped position.

[0068] Further to the above, the second jaw **10420** is pivotably coupled to the first jaw **10410** about a pivot axis. In various embodiments, the second jaw can both translate and rotate as it is being moved into its clamped position. In various alternative embodiments, a surgical instrument comprises a staple cartridge jaw that is movable between an unclamped position and a clamped position relative to an anvil jaw. In any event, the handle **10100** comprises a lock configured to releasably hold the closure actuator **10140** in its clamped position. The handle **10100** further comprises release actuators **10180b** on opposite sides thereof which, when actuated, unlock the closure actuator **10140** such that the end effector **10400** can be re-opened. In various alternative embodiments, the handle **10100** comprises an electric motor configured to move the closure tube **10240** proximally and/or distally when actuated by the clinician.

[0069] The end effector **10400** is attached to the shaft **10200** about an articulation joint **10500** and is rotatable within a plane about an articulation axis. The shaft **10200** defines a longitudinal axis and the end effector **10400** is articulatable between an unarticulated position in which the end effector **10400** is aligned with the longitudinal axis and articulated positions in which the end effector **10400** extends at a transverse angle relative to the longitudinal axis. In various embodiments, the surgical instrument **10000** comprises a first articulation joint which permits the end effector **10400** to be articulated in a first plane and a second articulation joint which permits the end effector **10400** to be articulated in a second plane which is orthogonal to the first plane, for example. The handle **10100** comprises at least one electric motor and a control system configured to control the operation of the electric motor in response to articulation actuators **10160** and **10170**. The electric motor comprises a brushless DC motor; however, the electric motor can comprise any suitable motor, such as a brushed DC motor, for example.

[0070] The entire disclosure of U.S. Pat. No. 10,149,683, entitled POWERED SURGICAL CUTTING AND STAPLING APPARATUS WITH MANUALLY RETRACTABLE FIRING SYSTEM, which issued on Dec. 11, 2018, is incorporated by reference herein. The entire disclosure of U.S. Patent Application Publication No. 2018/0125481, entitled MOTOR-DRIVEN SURGICAL CUTTING INSTRUMENT, which published on May 10, 2018, is incorporated by reference herein. The handle **10100** further comprises a replaceable and/or rechargeable battery **10300** attachable to the handle housing which powers the surgical instrument **10000**. The entire disclosure of U.S. Pat. No. 8,632,525, entitled POWER CONTROL ARRANGEMENTS FOR SURGICAL INSTRUMENTS AND BATTERIES, which issued on Jan. 21, 2014, is incorporated by reference herein.

[0071] Further to the above, the shaft **10200** is rotatable about a longitudinal axis extending through the shaft **10200**. The shaft **10200** is rotatably connected to the handle **10100** about a rotation joint **10220** and the shaft **10200** comprises one or more finger grooves defined therein which facilitate a clinician using the stapling instrument **10000** to rotate the shaft **10200**. In various embodiments, the surgical instrument **10000** comprises an electric motor and a rotation actuator that, when actuated by the clinician, powers the electric motor to rotate the shaft **10200** in a first direction or a second direction depending on the direction in which the rotation actuator is actuated.

[0072] Further to the above, the surgical instrument **10000** comprises a staple firing drive configured to eject the staples out of the staple cartridge. The staple firing drive comprises an electric motor and a firing member which is driven distally through a staple firing stroke by the electric motor. During the staple firing stroke, the firing member pushes the sled in the staple cartridge distally to eject the staples from the staple cartridge. The entire disclosure of U.S. Pat. No. 9,629,629, entitled CONTROL SYSTEMS FOR SURGICAL INSTRUMENTS, which issued on Apr. 25, 2017, is incorporated by reference herein.

[0073] The surgical instrument systems described herein are motivated by an electric motor; however, the surgical instrument systems described herein can be motivated in any suitable manner. In certain instances, the motors disclosed herein may comprise a portion or portions of a robotically controlled system. U.S. patent application Ser. No. 13/118,241, entitled SURGICAL STAPLING INSTRUMENTS WITH ROTATABLE STAPLE DEPLOYMENT ARRANGEMENTS, now U.S. Pat. No. 9,072,535, for example, discloses several examples of a robotic surgical instrument system in greater detail, the entire disclosure of which is incorporated by reference herein. The disclosures of International Patent Publication No. WO 2017/083125, entitled STAPLER WITH COMPOSITE CARDAN AND SCREW DRIVE, published May 18, 2017, International Patent Publication No. WO 2017/083126, entitled STAPLE PUSHER WITH LOST MOTION BETWEEN RAMPS, published May 18, 2017, International Patent Publication No. WO 2015/153642, entitled SURGICAL INSTRUMENT WITH SHIFTABLE TRANSMISSION, published Oct. 8, 2015, U.S. Patent Application Publication No. 2017/0265954, filed Mar. 17, 2017, entitled STAPLER WITH CABLE-DRIVEN ADVANCEABLE CLAMPING ELEMENT AND DUAL DISTAL PULLEYS, now U.S. Pat. No. 10,350,016, U.S. Patent Application Publication No. 2017/0265865, filed Feb. 15, 2017, entitled STAPLER WITH CABLE-DRIVEN ADVANCEABLE CLAMPING ELEMENT AND DISTAL PULLEY, now U.S. Pat. No. 10,631,858, and U.S. Patent Application Publication No. 2017/0290586, entitled STAPLING CARTRIDGE, filed on Mar. 29, 2017, now U.S. Pat. No. 10,722,233, are incorporated herein by reference in their entireties.

[0074] Various embodiments disclosed herein may be employed in connection with a robotic surgical system, such as the robotic system **1000** depicted in FIGS. 1-3, for example. FIG. 1 depicts a master controller **5001** that may be used in connection with a robotic arm cart **5100** depicted in FIG. 2. The master controller **5001** and the robotic arm cart **5100**, as well as their respective components and control systems, are collectively referred to herein as a robotic system **5000**. Examples of such systems and devices are disclosed in U.S. Pat. No. 7,524,320, entitled MECHANICAL ACTUATOR INTERFACE SYSTEM FOR ROBOTIC SURGICAL TOOLS, as well as U.S. Pat. No. 9,072,535, entitled SURGICAL STAPLING INSTRUMENTS WITH ROTATABLE STAPLE DEPLOYMENT ARRANGEMENTS, which are each hereby incorporated by reference herein in their respective entireties. The details of such systems and devices are not repeated herein for the sake of brevity. The master controller **5001** includes controls **5003** which are grasped and manipulated by the surgeon while the surgeon views the patient via a display **1002**. The controls **5003** can comprise manual input devices which move with multiple degrees of freedom, for example, and can further comprise an actuatable trigger for actuating surgical instruments, or tools, to close grasping jaws, staple and incise tissue, and/or apply an electrical potential to an electrode, for example.

[0075] With reference to FIGS. 2 and 3, the robotic arm cart **5100** is configured to actuate one or more surgical instruments, such as surgical instruments **6000**, for example, in response to inputs from the master controller **5001**. In various forms, the robotic arm cart **5100** includes a base **5002**, arm linkages including set-up joints **5104**, and instrument manipulators **5106**. Such an arrangement can facilitate the rotation of a surgical instrument **6000** around a point in space, which is described in U.S. Pat. No. 5,817,084, entitled REMOTE CENTER POSITIONING DEVICE WITH FLEXIBLE DRIVE, the entire disclosure of which is hereby incorporated by reference herein. This arrangement provides for pivoting rotation of a surgical instrument **6000** about an axis **5112a**, or

pitch axis. The arrangement also provides for rotation of the surgical instrument **6000** about an axis **5112b**, or yaw axis. The pitch and yaw axes **5112a**, **5112b** intersect at a remote center **5114**, which is aligned along an elongate shaft of the surgical instrument **6000**. A surgical instrument **6000** may have further degrees of driven freedom, including sliding motion along a longitudinal axis LT-LT. As the surgical instrument **6000** slides along the longitudinal axis LT-LT relative to the instrument manipulator **5106** (arrow **5112c**), the remote center **5114** remains fixed relative to a base **5116** of the instrument manipulator **5106**. To move the remote center **5114**, linkage **5108** is driven by one or more motors **5120** which move the linkage **5108** in response to commands from the master controller **5001** to position and/or manipulate the surgical instrument **6000** within the surgical site. Various other arrangements are disclosed in U.S. Pat. No. 5,878,193, entitled AUTOMATED ENDOSCOPE SYSTEM FOR OPTIMAL POSITIONING, the entire disclosure of which is hereby incorporated by reference herein.

[0076] Additionally, while the data communication between a robotic component and the processor of the robotic surgical system is primarily described herein with reference to communication between a surgical instrument, or tool, and the master controller **5001**, it should be understood that similar communication may take place between the circuitry of a manipulator, a set-up joint, an endoscope or other image capture device, or the like, and the processor of the robotic surgical system for component compatibility verification, component-type identification, component calibration (such as off-set or the like) communication, confirmation of coupling of the component to the robotic surgical system, or the like. In accordance with at least one aspect, various surgical instruments disclosed herein may be used in connection with other robotically-controlled or automated surgical systems and are not necessarily limited to use with the specific robotic system components shown in FIGS. **1-3** and described in the aforementioned references. Various robotic surgery systems and methods are disclosed in U.S. Pat. No. 6,132,368, entitled MULTI-COMPONENT TELEPRESENCE SYSTEM AND METHOD, the entire disclosure of which is hereby incorporated by reference herein.

[0077] Referring to FIG. **5**, a staple cartridge, such as staple cartridge **14000**, for example, comprises a cartridge body **11100** and an electronic circuit **11500** including sensors **11600**. The staple cartridge **14000** is similar to the other staple cartridges disclosed herein in many respects and such respects are not discussed herein for the sake of brevity. As discussed above, the cartridge body **11100** comprises a deck **11130** and longitudinal rows of staple cavities **11140** defined in the deck **11130**. Each staple cavity **11140** comprises a staple stored therein that is driven upwardly out of the staple cavity **11140** by a staple driver during a staple firing stroke. Each staple comprises a base and two legs extending from the base such that the legs extend generally upwardly and outwardly to form a V-shape configuration. In various instances, the legs of the staple are resiliently deflected inwardly by the proximal and distal end walls of the staple cavity **11140** when the staple is stored in the staple cavity **11140**. When the staple is driven upwardly out of the staple cavity **11140**, the legs of the staple emerge from the staple cavity **11140** and extend above the deck **11130** while the rest of the staple is pushed upwardly out of the staple cavity **11140**. The cartridge body **11100** comprises projections **11132** (FIG. **5B**) extending from the deck **11130** which are configured to guide and/or control the legs of the staples as the staples are being ejected from the staple cavities **11140**. A projection **11132** is positioned at the distal end of each staple cavity **11140** and at the proximal end of each staple cavity **11140**. However, alternative embodiments are envisioned in which a projection **11132** is positioned at only one end of each staple cavity **11140**. Moreover, various embodiments are envisioned in which some of the staple cavities **11140** do not comprise projections **11132** at the ends thereof. The projections **11132** are further configured to engage the patient tissue positioned against the deck **11130** and limit the flow or movement of the patient tissue relative to the deck **11130**.

[0078] In various embodiments, the electronic circuit **11500** comprises a substrate including features engaged with the projections **11132**. In at least one embodiment, the substrate comprises

apertures defined therein, the sidewalls of which are engaged with the projections **11132**. The apertures are in a snap-fit and/or press-fit arrangement with the projections **11132** such that the electronic circuit **11500** is held in position relative to the cartridge body **11100**. In at least one embodiment, the projections **11132** comprise at least partially annular or circumferential shoulders which hold the sensor circuit **11500** against the cartridge body **11100**.

[0079] In various embodiments, a sensor circuit of a staple cartridge is comprised of a conductive material printed on the deck of the cartridge body. In at least one embodiment, the conductive material is comprised of metal particles bonded to the deck which form an electrical circuit connecting the sensors. In at least one such embodiment, the printed electrical circuit is printed onto the cartridge body with a three-dimensional printer. In various embodiments, the sensor circuit comprises electrodes, or contacts, that are printed onto the cartridge body. In at least one embodiment, the sensor circuit comprises electrodes which comprise a polygonal surface configured to contact the tissue. In at least one alternative embodiment, the electrodes comprise a curved and/or tortuous path on the deck surface which, in various instances, can increase the contact area between the electrodes and the tissue. In at least one embodiment, the electrodes comprise needles extending therefrom which are configured to penetrate the tissue. In at least one embodiment, the needles comprise a diameter of about 1 μm , for example. In various instances, the needles provide parallel signal paths between the tissue and the sensor circuit within one electrode to improve the sensitivity of the sensor circuit. In at least one embodiment, a conductive grease or conductive viscous agent covers the tissue contact points of the sensor circuit which improves the contact between the electrodes and the tissue. In various embodiments, portions of the sensor circuit are embedded in the cartridge body. In at least one such embodiment, the sensor circuit comprises flat, thin conductors that are embedded into the cartridge body when a plastic material, for example, is overmolded onto portions of the conductors. Portions of the conductors, however, remain exposed to provide tissue engaging pads and/or electrically-conductive attachment points for soldering sensors thereto. In at least one embodiment, part of the cartridge sensor circuit can be defined on the lateral sidewalls of the cartridge jaw. In at least one such embodiment, a proximal portion and a distal portion of the sensor circuit are defined on the cartridge body and an intermediate portion of the sensor circuit is defined on the cartridge jaw that electrically connects the proximal portion and the distal portion of the sensor circuit. In at least one embodiment, the portions of the sensor circuit mounted to the cartridge jaw comprise conductive strips mounted to the sidewalls. When the staple cartridge is seated in the cartridge jaw, the cartridge sensor circuit engages the conductive strips to complete the circuit.

[0080] As discussed above, a sensor circuit can include conductive tissue-contacting surfaces. In various embodiments, a sensor circuit can include non-conductive tissue-contacting surfaces. In at least one embodiment, a sensor circuit comprises one or more capacitive electrodes. In various instances, projected capacitance measurement techniques are used to measure the presence of the tissue over the capacitive electrodes and/or a property of the tissue over the capacitive electrodes. In at least one embodiment, each capacitive electrode comprises an insulative covering which covers capacitive pads contained therein. In various instances, further to the above, surface capacitance measurement techniques can be used. In various embodiments, a sensor circuit comprises one or more inductive sensors. In at least one embodiment, an eddy current is induced in each of the inductive sensors which changes when the tissue contacts the sensors. In such embodiments, the changes to the sensor eddy currents are detected by the control system of the staple cartridge. In various embodiments, the sensor circuit can comprise temperature sensors which are used to detect the presence of tissue over the temperature sensors. In at least one embodiment, the sensor circuit comprises electrodes comprised of a doped polycrystalline ceramic comprising barium titanate (BaTiO_3), for example. The resistance of these ceramic materials changes in response to temperature changes, such as when patient tissue is positioned against the electrodes. The cartridge processor is configured to employ an algorithm to monitor the resistance

fluctuations in the ceramic materials to assess whether or not tissue was positioned against the electrodes. In various instances, the electrodes of the sensor circuit are in a parallel arrangement such that a detected resistance, capacitance, voltage, and/or current change can be directly related to the position of a sensor. With this information, the processor can assess whether and where tissue is positioned over the staple cartridge.

[0081] Referring to FIGS. 5A and 5D, the staple cartridge **14000** further comprises a laminate material **14900** mounted to one or more components of the staple cartridge **14000** to control the electrical effects created within the cartridge components by the fields emitted from and/or surrounding the staple cartridge **14000**. In at least one instance, the laminate material **14900** comprises a flux field directional material including at least two layers—a first layer **14910**, or cover, and a second layer **14920** of magnetic material attached to the first layer **14910**. The first layer **14910** is comprised of polyethylene terephthalate, for example, which protects the second layer **14920**, but can be comprised of any suitable material. The second layer **14920** is comprised of a sintered ferrite sheet, for example, but can be comprised of any suitable material. In at least one instance, an adhesive layer **14930** comprised of a pressure-sensitive adhesive, for example, is bonded to the second layer **14920** and is used to attach the laminate material **14900** to one or more components of the staple cartridge **14000**, as discussed further below. In at least one instance, the laminate material **14900** is a Flux Field Directional Material EM15TF manufactured by 3M, for example.

[0082] In various embodiments, further to the above, laminate material **14900** is bonded to the cartridge body **11100** and is arranged to change and/or control the shape of the fields extending from the cartridge antennas. In at least one embodiment, the laminate material **14900** focuses the fields away from the metal cartridge jaw of the surgical instrument **10000** in which the staple cartridge **14000** is seated. In at least one instance, the cartridge body **11100** is comprised of plastic and the laminate material **14900** is mounted to the cartridge body **11100** such that the laminate material **14900** surrounds, or at least substantially surrounds, the cartridge antennas. In at least one instance, laminate material **14900** is mounted to the cartridge body **11100** at a location which is intermediate the cartridge data coil **11540"** and the cartridge power coil **11545"** such that the cartridge coils **11540"** and **11545"** are separated by the laminate material **14900**. In various embodiments, laminate material **14900** is bonded to the metal walls of the cartridge jaw **10410**. In at least one instance, laminate material **14900** is mounted to the metal walls of the cartridge jaw **10410** at a location which is intermediate the instrument data coil **10540"** and the power transmission coil **10545"**. In various embodiments, the laminate material **14900** bonds the cartridge data antenna **11530"** and/or the cartridge power antenna **11535"** to the cartridge body **11100**. In at least one embodiment, the laminate material **14900** bonds the instrument data antenna **10530"** and/or the instrument power antenna **10535"** to the metal cartridge jaw **10410**.

[0083] In various embodiments, further to the above, laminate material **14900** is mounted to the metal pan **11700**. In at least one such instance, laminate material **14900** is positioned intermediate the metal pan **11700** and the cartridge data antenna **11530"** and, also, intermediate the metal pan **11700** and the cartridge power antenna **11535"**. Such an arrangement can focus the fields created by the antennas **11530"** and **11535"** away from the metal pan **11700** to minimize the electrical effects that the fields have on the metal pan **11700**. In various embodiments, laminate material **14900** is mounted to the movable components of the staple cartridge **14000**. In at least one instance, referring to FIG. 5D, laminate material **14900** is mounted to the sled **11400**. In at least one such instance, laminate material **14900** is mounted to the lateral sides **11410** of the sled **11400**, for example. In at least one instance, referring to FIG. 5A, laminate material **14900** is mounted to one or more of the staple drivers **11300**, for example. In at least one such instance, laminate material **14900** is mounted to the lateral sides **11310** of the staple drivers **11300**. Laminate material **14900** can be mounted to all of the staple drivers **11300**, or just the staple drivers **11300** adjacent the cartridge antennas **11530"** and **11535"**, for example.

[0084] Further to the above, the fields generated by the cartridge antennas and/or instrument antennas can affect the output of the sensors **11600**. Such an effect can be reduced or mitigated by the laminate material **14900**, for example. In various instances, the processor of the staple cartridge **14000** is configured to electronically account for the effect that the antenna fields will have on the sensors **11600**. In at least one such instance, the cartridge processor can monitor when signals are being transmitted between the antenna couples and, in such instances, modify the sensor outputs being received from the sensors **11600** before transmitting the sensor outputs to the surgical instrument processor and/or recording the sensor outputs in a memory device in the staple cartridge **14000**. When signals are not being transmitted between the antenna couples, the sensor outputs may not need to be modified by the processor before being transmitted to the surgical instrument processor and/or recorded in a memory device in the staple cartridge **14000**. In various instances, the processor can apply a first compensation factor to the sensor outputs when the power antenna couple is transmitting signals, a second compensation factor to the sensor outputs when the signal antenna couple is transmitting signals, and a third compensation factor to the sensor outputs when both antennas are transmitting signals. In at least one such instance, the third compensation factor is larger than the first compensation factor and the first compensation factor is larger than the second compensation factor, for example.

[0085] Further to the above, the circuit **11500** is flush with the top surface of the deck **11130** and/or recessed with respect to the top surface of the deck **11130**. In various instances, the staple cartridge **11000** further comprises latches rotatably mounted thereto which are rotatable from an unlatched position to a latched position to hold the circuit **11500** in the circuit slot **11160**. The latches engage the cartridge body **11100** in a press-fit and/or snap-fit manner when the latches are in their latched position. When the latches are in their latched position, the latches are flush with and/or recessed below the top surface of the deck **11130**. In at least one embodiment, the projections **11132** are mounted to and/or integrally-formed with the latches and/or any other suitable restraining features. In any event, the circuit **11500** comprises one or more sensors which are held in place relative to the cartridge body **11100** as a result of the above.

[0086] FIG. 6 is a logic flow diagram of an algorithm **1190** depicting a control program or a logic configuration for modulating a sensor parameter of the sensor array **1036**, in accordance with at least one aspect of the present disclosure. In the illustrated example, the algorithm **1190** includes detecting **1191** a tissue contact status of the staple cartridge **1046**. The algorithm **1190** further includes selectively modulating **1182** a sensor parameter of one or more sensors of the sensor array **1036** in accordance with the detected tissue contact status. In the illustrated example, the algorithm **1190** is implemented, or at least partially implemented, by the control circuit **1026**. In other examples, various aspects of the algorithm **1190** can be implemented by other control circuits such as, for example, the control circuit **1049**, or any other suitable control circuit. For brevity the following description will focus on executing various aspects of the algorithm **1190** by the control circuit **1026**.

[0087] In various aspects, detecting **1191** the tissue contact status of the staple cartridge **1046** is performed at each of a plurality of closure states. As the closure of the end effector **1040** commences, the size and/or position of the tissue in contact with the sensor array **1036** of the staple cartridge **1046** may change. To optimize sensor data collection, transmission, and/or processing, the control circuit **1026** can be configured to adjust one or more sensor parameters of one or more sensors, or groups of sensors, of the sensor array **1036** based on whether tissue contact is detected at the different closure states.

[0088] In certain exemplifications, as illustrated in FIG. 7, the sensor array **1036** is disposed along a length L of the staple cartridge **1046**. However, the tissue grasped by the end effector **1040** may cover a region **1193** extending only along a portion of the length L, for example extending along a length L1. In such instances, sensor data from sensors beyond the region **1193** can be assigned a lower priority than sensor data from sensors within the region **1193**. A control circuit **1026** can be

configured to determine a priority level of the sensors of the sensor array **1036** based on their location with respect to the region **1193**, for example. Furthermore, the control circuit **1026** can be configured to switch sensors of the sensor array **1036** that are within the region **1193** to an active mode **1083** and/or switch sensors of the sensor array **1136** that are outside the region **1193** to an idler mode **1084**, for example.

[0089] In various aspects, tissue contact detection can be accomplished by a tissue contact circuit **2830**, as described in greater detail elsewhere in the present disclosure. The tissue contact circuit **2830** is in open circuit mode with no tissue located against the sensors **2788a**, **2788b**. The tissue contact circuit **2830** is transitioned to a closed circuit mode by the tissue **2820**. The sensors **2788a**, **2788b** are powered by voltage source **V** and a sensors circuit **2790** measures a signal generated by the sensors **2788a**, **2788b**. In some aspects, the sensors **2788a**, **2788b** may include a pair of opposing electrode plates to make electrical contact with the tissue **2820**.

[0090] Any of the sensors **2788a**, **2788b** disclosed herein may include, and are not limited to, electrical contacts placed on an inner surface of a jaw which, when in contact with tissue, close a sensing circuit that is otherwise open. The contact sensors may also include sensitive force transducers that detect when the tissue being clamped first resists compression. Force transducers may include, and are not limited to, piezoelectric elements, piezoresistive elements, metal film or semiconductor strain gauges, inductive pressure sensors, capacitive pressure sensors, and resistive sensors.

[0091] Further to the above, a control circuit **1026**, for example, may receive one or more signals from the sensor circuit **2790** and/or sensors **2788a**, **2788b** indicative of a tissue contact status of one or more regions along the length **L** of the staple cartridge **1046**. In response, the adjust one or more sensor parameters of one or more sensors, or groups of sensors, the control circuit **1026** can be configured to adjust sensor parameters of one or more sensors of the sensor array **1036** in the one or more regions based on the tissue contact status.

[0092] Additional details are disclosed in U.S. Pat. No. 10,595,887, titled SYSTEMS FOR ADJUSTING END EFFECTOR PARAMETERS BASED ON PERIOPERATIVE INFORMATION, and issued Mar. 24, 2020, U.S. Pat. No. 9,724,094, titled ADJUNCT WITH INTEGRATED SENSORS TO QUANTIFY TISSUE COMPRESSION, and issued Aug. 8, 2017, and U.S. Pat. No. 9,808,246, titled METHOD OF OPERATING A POWERED SURGICAL INSTRUMENT, and issued Nov. 7, 2017, the entireties of disclosures of which are hereby incorporated by reference herein.

[0093] In one general aspect, the present disclosure provides methods of monitoring multiple sensors over time to detect moving characteristics of tissue located in the jaws of the end effector. In one aspect, the end effector comprises a cartridge. More than one sensor can be located on a cartridge to sense the motion of the tissue from one sensor towards an adjacent sensor. In a stapling cartridge, multiple sensors may be located on the stapling cartridge to sense movement of tissue by monitoring a property of the tissue. In one aspect, the tissue property could be an electrical property of the tissue such as impedance or capacitance. In another aspect, monitoring the impedance of the tissue from one time point to the next can allow the system to detect the motion of the tissue from one sensor towards the next.

[0094] In one aspect, a method of monitoring multiple sensors over time to detect moving characteristics of the tissue comprises monitoring multiple sensors over time to detect tissue movement relative to at least two sensed locations. The method provides real-time tissue flow sensing through monitoring a sensed tissue property through time.

[0095] Turning now to FIG. **8**, which illustrates a diagram of a surgical instrument **2750** comprising an instrument housing **2800** and an end effector **2752** inductively coupled to the instrument housing **2800** via a set of coils **2818** implementing a wireless power and data communication system, in accordance with at least one aspect of the present disclosure. The instrument housing **2800** comprises an energy source **2762** and a control circuit **2760** inductively

coupled to the end effector **2752**. Power from the energy source **2762** is inductively coupled to the end effector **2752** from a primary coil **2802** tuned for power located in the instrument housing **2800** to a secondary coil **2804** tuned for power located in the end effector **2752**. Data is transmitted between the control circuit **2760** and the end effector sensor circuits **2790** between a primary coil **2816** tuned for data located in the instrument housing **2800** and a secondary coil **2814** tuned for data located in the end effector **2752**.

[0096] FIG. **8** illustrates one implementation of the transmission system **1045** for wireless transmission of power and data. In the implementation illustrated in FIG. **8**, power and data are transmitted separately. In other implementations, as described supra, power and data are transmitted sequentially or simultaneously. For brevity, the following description focuses on the implementation of the transmission system **1045** that is configured to separately transmit power and data. However, it is understood that the other implementations of the transmission system **1045** can be equally utilized.

[0097] In various aspects, the end effector **2752** comprises a cartridge **2768** and an anvil **2766** pivotally coupled to the cartridge **2768**. A plurality of sensors **2788** may be disposed in the cartridge **2768**, the anvil **2766**, or both. As described supra, the end effector **2752** comprises secondary coils **2804**, **2814** to receive power from the instrument housing **2800** and communicate between the end effector **2752** circuits and the instrument housing **2800** circuits, respectively. Power from the secondary coil **2804** is rectified by a rectifier circuit **2806** and filter capacitor **2808** and is provided to a plurality of sensors **2788** via an analog multiplexer **2810** or other analog switching circuit. Signals from the sensors **2788** are transmitted through the analog multiplexer **2810**, coupled to a near field communication (NFC) tag **2812**, and coupled to the control circuit **2760** from the secondary coil **2814** located in the end effector **2752** and the primary coil **2816** located in the instrument housing **2800**. The NFC tag **2812** is configured to transmit data from the cartridge **2768**. The sensors **2788** may be configured to measure tissue impedance, tissue temperature, tissue capacitance, tissue inductance, elapsed time, among other tissue parameters explained in the following description.

[0098] In other aspects, the cartridge **2768** portion of the end effector **2752** may comprise electrodes to receive electrosurgical energy to assist or enhance the tissue sealing process. In such aspects, some or all of the plurality of sensors **2788** may act as electrodes to deliver the electrosurgical energy through the tissue clamped between the anvil **2766** and the cartridge **2768**. In such aspects, the plurality of sensors **2788** may be configured to measure tissue parameters such as impedance, capacitance, among other tissue parameters explained in the following description.

[0099] In other aspects, the end effector **2752** may comprise a clamp arm assembly and an ultrasonic blade for cutting and sealing tissue clamped between the clamp arm assembly and the ultrasonic blade instead of the anvil **2766** and cartridge **2768** as shown in the example of FIG. **8**. In such aspects comprising a clamp arm assembly and ultrasonic blade, the plurality of sensors **2788** may be disposed in the clamp arm assembly and the electrical return path may be provided through the electrically conductive ultrasonic blade. The plurality of sensors **788** may be configured to measure tissue parameters such as impedance, capacitance, among other tissue parameters explained in the following description.

[0100] In other aspects, the end effector **2752** may comprise a pair of jaws configured with electrodes to deliver electrosurgical energy to seal tissue clamped between the jaws instead of the anvil **2766** and cartridge **2768** as shown in the example of FIG. **8**. One of the jaws may be configured with a knife slot for cutting through the tissue after sealing. In such aspects, the plurality of sensors **2788** may be disposed in either jaw or both. The plurality of sensors **2788** may be configured to measure tissue parameters such as impedance, capacitance, among other tissue parameters explained in the following description.

[0101] In other aspects, the end effector **2752** may comprise a clamp arm assembly and an ultrasonic blade instead of the anvil **2766** and cartridge **2768** as shown in the example of FIG. **8**. In

such aspects, the clamp arm assembly is configured with electrodes for receiving electrosurgical energy for sealing tissue located between the clamp arm assembly and the ultrasonic blade. The electrical return path for the electrosurgical energy is provided through the electrically conductive ultrasonic blade. In such aspects, the ultrasonic blade is utilized to cut the sealed tissue clamped between the clamp arm assembly and the ultrasonic blade. The plurality of sensors **2788** may be configured to measure tissue parameters such as impedance, capacitance, among other tissue parameters explained in the following description.

[0102] In certain instances, as described in greater detail elsewhere in the present disclosure, wireless power and/or data transmission between an instrument housing **2800** and the end effector **2752** encompasses a wireless power and/or data transmission between the surgical instrument **2750** and the staple cartridge **2768**. For example, the primary coils **2802**, **2816** can be disposed on a cartridge channel of the end effector **2752**, and the secondary coils **2804**, **2814** can be disposed on the staple cartridge **2768** such that the primary coils **2802**, **2816** and the secondary coils **2804**, **2814** are aligned for a wireless connection when the staple cartridge **2768** is seated in the cartridge channel. In such instances, the instrument housing **2800** may encompass a proximal housing including the energy source **2762** and the control circuit **2760**, a shaft extending distally from the proximal housing, and the cartridge channel.

[0103] FIG. **9** illustrates a block diagram of the surgical instrument **2750** shown in FIG. **8** comprising an instrument housing **2800** and an end effector **2752** inductively coupled to the instrument housing **2800** via a set of coils **2818** implementing a wireless power and data communication system, in accordance with at least one aspect of the present disclosure. In one aspect, the surgical instrument **2750** is configured or programmed to control the distal translation of a displacement member such as the I-beam **2764**. The surgical instrument **2750** comprises an end effector **2752** that may comprise an anvil **2766**, an I-beam **2764** (including a sharp cutting edge), and a removable cartridge **2768**. The end effector **2752** comprises sensors **2788** and a sensors circuit **2790** coupled to the sensors **2788**. Power is inductively coupled to the sensor circuit **2790** and to the sensors **2788** through coils **2802**, **2804** via near field communication. Signals (e.g., voltage, current, resistance, impedance, capacitance, inductance, frequency, phase, etc.) from the sensors **2788** are conditioned by the sensors circuit **2790**. The signals or data corresponding to the signals are communicated between the sensors circuit **2790** in the end effector **2752** and the control circuit **2760** in the instrument housing **2800** via near field communication inductive coupling between the coils **2814**, **2816**.

[0104] It will be appreciated that the sensors **2788** may be located in any suitable location in the end effector **2752**. In one aspect, the sensors **2788** are arranged in an array in the cartridge **2768**. In another aspect, the sensors **2788** are arranged in an array in the anvil **2766**. In various aspects, the sensors **2788** are arranged in arrays in the cartridge **2768** and the anvil **2766**. The control circuit **2760** may be configured to monitor the sensors **2788** over time to detect moving characteristics of tissue located in the jaws of the end effector **2752**. In one aspect, the jaws of the end effector **2752** may be comprised of the anvil **2766** and the cartridge **2768**, for example.

[0105] The position, movement, displacement, and/or translation of a linear displacement member, such as the I-beam **2764**, can be measured by an absolute positioning system, sensor arrangement, and position sensor **2784**. A control circuit **2760** may be configured or programmed to control the translation of the displacement member, such as the I-beam **2764**. The control circuit **2760**, in some examples, may comprise one or more microcontrollers, microprocessors, or other suitable processors for executing instructions that cause the processor or processors to control the displacement member, e.g., the I-beam **2764**. In other aspects, the control circuit **2760** may comprise analog or digital circuits such as, for example, programmable logic devices (PLD), field programmable gate arrays (FPGA), discrete logic, or other hardware circuits, software, and/or firmware, or other machine executable instructions to perform the functions explained in the following description.

[0106] In one aspect, the control circuit **2760** may be configured or programmed to sense multiple longitudinal and lateral locations within the end effector **2752** independently and to use these different sensed locations with a localized predetermined return path to sense changes in the impedance of tissue grasped between the anvil **2766** and the cartridge **2768** both laterally and longitudinally to be able to detect any specific tissue mid-thickness measure by triangulating at least two interconnected session combinations. For example, the sensors **2788** may comprise an array of impedance sensors distributed laterally and longitudinally along the length of the stapler jaws, i.e., the cartridge **2768** and anvil **2766**. As the jaws are closing, the control circuit **2760** may track the local impedance over time during the course of the jaw closure for each sensor, based on readings from the timer/counter **2781**, or using software timing techniques. This time history can be used to infer, if present, regions of heterogeneous impedance values—where there are distinct changes or anomalies that mark a particular location. These baseline location(s) are noted and tracked as firing is initiated. Once initiated, the position histories of these locations is tracked and used for feedback control of the firing process. In another example, the control circuit may be configured or programmed to modify functions of the surgical instrument **2750** to alter tissue flow during firing of the I-beam **2764** including changing the firing speed, pauses (complete stops) in firing, closure force, among other parameters.

[0107] In other aspects, the control circuit **2760** may be configured or programmed to predict an amount of tissue flow occurring in the jaws of the end effector **2752** by monitoring the sensors **2788**. Knowledge of tissue type from situational awareness and/or other device sensed measures, e.g., rate of change of closure load during closure, rate of change of closure load after closure is complete, etc. can be used by the control circuit **2760** to predict tissue flow. Accordingly, in one aspect, the control circuit **2760** is configured or programmed to determine tissue type or condition by combining tissue flow during jaw closure with force feedback of the anvil **2766** closure system.

[0108] In another example, the predictions can be further refined by using the sensors **2788** to measure tissue impedance, among other parameters, detect rigid or foreign objects in the jaws, measure magnitude of tissue impedance, measure tissue flow during jaw closure, etc. In another example, the control circuit **2760** may execute a jaw closure algorithm to sense tissue movements during closure as an indicator of the potential effect of each change during firing of the I-beam **2764**. For example, at a first closure rate, the control circuit **2760** estimates the magnitude/direction of tissue flow, adjusts the closure rate of the jaws, and observes or records the changes in tissue flow within the jaws. In another example, the control circuit **2760** may be configured or programmed to predict post-fire tissue position by utilizing closure flow in combination with closure force feedback prior to firing to provide feedback to surgeon and allowing an opportunity to reposition the end effector **2752** to ensure tissue is fully captured in cut the line of the end effector **2752**.

[0109] In other aspects, the control circuit **2760** may be configured or programmed to receive data for various configurations of the sensors **2788** to monitor and interrogate tissue. This may include, monitoring tissue impedance, and tracking the impedance of the tissue across a single electrode or segmented electrode set configured along the length of the cartridge **2788**. The control circuit **2760** may be configured or programmed to monitor spectrographic impedance by utilizing sweeps of different frequencies and monitoring the tissue impedance to the power and frequency to determine the physiological composition of the tissue, monitoring capacitance of the tissue, and determining the tissue characteristics and gap relationship of the jaws to determine the amount of tissue present within the jaws. In another aspect, the control circuit **2760** may be configured or programmed to measure light transmissivity, refractivity or Doppler effects to determine tissue characteristics. Local light refractivity analysis may be employed to determine the surface conditions of the tissue to monitor irregularities within the tissue captured between the jaws. The control circuit **2760** may be configured or programmed to monitor local moving particles of tissue using Doppler effect frequency analysis of the light.

[0110] In one aspect, a timer/counter **2781** provides an output signal, such as the elapsed time or a digital count, to the control circuit **2760** to correlate the position of the I-beam **2764** as determined by the position sensor **2784** with the output of the timer/counter **2781** such that the control circuit **2760** can determine the position of the I-beam **2764** at a specific time (t) relative to a starting position. The timer/counter **2781** may be configured to measure elapsed time, count external events, or time external events. In other aspects, the timer/counter **2781** may be employed to measure elapsed time to monitor the sensors **2788** over time to detect moving characteristics of tissue located in the jaws of the end effector **2752**.

[0111] The control circuit **2760** may generate a motor set point signal **2772**. The motor set point signal **2772** may be provided to a motor controller **2758**. The motor controller **2758** may comprise one or more circuits configured to provide a motor drive signal **2774** to the motor **2754** to drive the motor **2754** as described herein. In some examples, the motor **2754** may be a brushed DC electric motor. For example, the velocity of the motor **2754** may be proportional to the motor drive signal **2774**. In some examples, the motor **2754** may be a brushless DC electric motor and the motor drive signal **2774** may comprise a PWM signal provided to one or more stator windings of the motor **2754**. Also, in some examples, the motor controller **2758** may be omitted, and the control circuit **2760** may generate the motor drive signal **2774** directly.

[0112] The motor **2754** may receive power from an energy source **2762**. The energy source **2762** may be or include a battery, a super capacitor, or any other suitable energy source. The motor **2754** may be mechanically coupled to the I-beam **2764** via a transmission **2756**. The transmission **2756** may include one or more gears or other linkage components to couple the motor **2754** to the I-beam **2764**. A position sensor **2784** may sense a position of the I-beam **2764**. The position sensor **2784** may be or include any type of sensor that is capable of generating position data that indicate a position of the I-beam **2764**. In some examples, the position sensor **2784** may include an encoder configured to provide a series of pulses to the control circuit **2760** as the I-beam **2764** translates distally and proximally. The control circuit **2760** may track the pulses to determine the position of the I-beam **2764**. Other suitable position sensors may be used, including, for example, a proximity sensor. Other types of position sensors may provide other signals indicating motion of the I-beam **2764**. Also, in some examples, the position sensor **2784** may be omitted. Where the motor **2754** is a stepper motor, the control circuit **2760** may track the position of the I-beam **2764** by aggregating the number and direction of steps that the motor **2754** has been instructed to execute. The position sensor **2784** may be located in the end effector **2752** or at any other portion of the instrument.

[0113] The control circuit **2760** may be in communication with one or more sensors **2788** located in the end effector **2752**. The sensors **2788** may be positioned in the end effector **2752** and adapted to operate with the surgical instrument **2750** to measure various derived parameters such as gap distance versus time, tissue compression versus time, anvil strain versus time, tissue movement versus time, tissue impedance, tissue capacitance, spectroscopic impedance, light transmissivity, refractivity or Doppler effects, among other parameters. The sensors **2788** may comprise a magnetic sensor, a magnetic field sensor, a strain gauge, a pressure sensor, a force sensor, an inductive sensor such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor for measuring one or more parameters of the end effector **2752**. The sensors **2788** may include one or more sensors.

[0114] The one or more sensors **2788** may comprise a strain gauge, such as a micro-strain gauge, configured to measure the magnitude of the strain in the anvil **2766** during a clamped condition. The strain gauge provides an electrical signal whose amplitude varies with the magnitude of the strain. The sensors **2788** may comprise a pressure sensor configured to detect a pressure generated by the presence of compressed tissue between the anvil **2766** and the cartridge **2768**. The sensors **2788** may be configured to detect impedance of a tissue section located between the anvil **2766** and the cartridge **2768** that is indicative of the thickness and/or fullness of tissue located therebetween.

[0115] The sensors **2788** may be configured to measure forces exerted on the anvil **2766** by a

closure drive system. For example, one or more sensors **2788** can be at an interaction point between a closure tube and the anvil **2766** to detect the closure forces applied by a closure tube to the anvil **2766**. The forces exerted on the anvil **2766** can be representative of the tissue compression experienced by the tissue section captured between the anvil **2766** and the cartridge **2768**. The one or more sensors **2788** can be positioned at various interaction points along the closure drive system to detect the closure forces applied to the anvil **2766** by the closure drive system. The one or more sensors **2788** may be sampled in real time during a clamping operation by a processor of the control circuit **2760**. The control circuit **2760** receives real-time sample measurements to provide and analyze time-based information and assess, in real time, closure forces applied to the anvil **2766**.

[0116] A current sensor **2786** can be employed to measure the current drawn by the motor **2754**. The force required to advance the I-beam **2764** corresponds to the current drawn by the motor **2754**. The force is converted to a digital signal and provided to the control circuit **2760**.

[0117] The drive system of the surgical instrument **2750** is configured to drive the displacement member, cutting member, or I-beam **2764**, by a brushed DC motor with gearbox and mechanical links to an articulation and/or knife system. Another example is the electric motor **2754** that operates the displacement member and the articulation driver, for example, of an interchangeable shaft assembly. An outside influence is an unmeasured, unpredictable influence of things like tissue, surrounding bodies and friction on the physical system. Such outside influence can be referred to as drag which acts in opposition to the electric motor **2754**. The outside influence, such as drag, may cause the operation of the physical system to deviate from a desired operation of the physical system.

[0118] Various example aspects are directed to a surgical instrument **2750** comprising an end effector **2752** with motor-driven surgical stapling and cutting implements. For example, a motor **2754** may drive a displacement member distally and proximally along a longitudinal axis of the end effector **2752**. The end effector **2752** may comprise a pivotable anvil **2766** and, when configured for use, a cartridge **2768** positioned opposite the anvil **2766**. A clinician may grasp tissue between the anvil **2766** and the cartridge **2768**, as described herein. When ready to use the instrument **2750**, the clinician may provide a firing signal, for example by depressing a trigger of the instrument **2750**. In response to the firing signal, the motor **2754** may drive the displacement member distally along the longitudinal axis of the end effector **2752** from a proximal stroke begin position to a stroke end position distal of the stroke begin position. As the displacement member translates distally, an I-beam **2764** with a cutting element positioned at a distal end, may cut the tissue between the cartridge **2768** and the anvil **2766**.

[0119] In various examples, the control circuit **2760** may be configured or programmed to control the distal translation of the displacement member, such as the I-beam **2764**, for example, based on one or more tissue conditions. The control circuit **2760** may be configured or programmed to sense tissue conditions, such as thickness, flow, impedance, capacitance, light transmissivity, either directly or indirectly, as described herein. The control circuit **2760** may be configured or programmed to select a firing control program based on tissue conditions. A firing control program may describe the distal motion of the displacement member. Different firing control programs may be selected to better treat different tissue conditions. For example, when thicker tissue is present, the control circuit **2760** may be configured or programmed to translate the displacement member at a lower velocity and/or with lower power. When thinner tissue is present, the control circuit **2760** may be configured or programmed to translate the displacement member at a higher velocity and/or with higher power.

[0120] The entire disclosures of U.S. Pat. No. 8,622,274, entitled MOTORIZED CUTTING AND FASTENING INSTRUMENT HAVING CONTROL CIRCUIT FOR OPTIMIZING BATTERY USAGE, U.S. Pat. No. 10,135,242, entitled SMART CARTRIDGE WAKE UP OPERATION AND DATA RETENTION, U.S. Pat. No. 10,548,504, entitled OVERLAID MULTI SENSOR RADIO

FREQUENCY (RF) ELECTRODE SYSTEM TO MEASURE TISSUE COMPRESSION, U.S. Pat. No. 9,993,248, entitled SMART SENSORS WITH LOCAL SIGNAL PROCESSING, U.S. Patent Application Publication No. 2016/0256071, entitled OVERLAID MULTI SENSOR RADIO FREQUENCY (RF) ELECTRODE SYSTEM TO MEASURE TISSUE COMPRESSION, now U.S. Pat. No. 10,548,504, U.S. Patent Application No. 2018/0168625, entitled SURGICAL STAPLING INSTRUMENTS WITH SMART STAPLE CARTRIDGES, U.S. Patent Application No. 2018/0250002, entitled POWERED SURGICAL DEVICES HAVING TISSUE SENSING FUNCTION, and International Publication No. WO 2018/049206, entitled STAPLER RELOAD DETECTION AND IDENTIFICATION, and U.S. patent application Ser. No. 16/354,470, entitled are incorporated by reference herein.

[0121] The entire disclosures of U.S. Pat. No. 4,785,180, titled OPTOELECTRIC SYSTEM HOUSED IN A PLASTIC SPHERE, issued Nov. 15, 1988, U.S. Pat. No. 6,804,012, titled ARRANGEMENT FOR THE DETECTION OF RELATIVE MOVEMENTS OR RELATIVE POSITION OF TWO OBJECTS, issued Oct. 12, 2004, European Patent Application No. 1,850,210, titled OPTOELECTRONIC DEVICE FOR DETERMINING RELATIVE MOVEMENTS OR RELATIVE POSITIONS OF TWO OBJECTS, published Oct. 31, 2007, U.S. Patent Application Publication No. 2008/0001919, titled USER INTERFACE DEVICE, published Jan. 3, 2008; and U.S. Pat. No. 7,516,675, titled JOYSTICK SENSOR APPARATUS, issued Apr. 14, 2009 are incorporated by reference herein. Generally, these references describe multi-dimensional input devices and/or sensor arrangements.

[0122] The entire disclosures of U.S. patent application Ser. No. 16/354,470 U.S., titled SEGMENTED CONTROL INPUTS FOR SURGICAL ROBOTIC SYSTEMS, filed Mar. 15, 2019, U.S. Pat. No. 4,785,180, titled OPTOELECTRIC SYSTEM HOUSED IN A PLASTIC SPHERE, issued Nov. 15, 1988, U.S. Pat. No. 6,804,012, titled ARRANGEMENT FOR THE DETECTION OF RELATIVE MOVEMENTS OR RELATIVE POSITION OF TWO OBJECTS, issued Oct. 12, 2004, European Patent Application No. 1,850,210, titled OPTOELECTRONIC DEVICE FOR DETERMINING RELATIVE MOVEMENTS OR RELATIVE POSITIONS OF TWO OBJECTS, published Oct. 31, 2007, U.S. Patent Application Publication No. 2008/0001919, titled USER INTERFACE DEVICE, published Jan. 3, 2008; U.S. Pat. No. 7,516,675, titled JOYSTICK SENSOR APPARATUS, issued Apr. 14, 2009, and U.S. patent application Ser. No. 15/940,627, titled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, filed Mar. 29, 2018, are hereby incorporated by reference in their entirety. Generally, these references describe robotic surgical systems and multi-dimensional input devices and/or sensor arrangements.

[0123] Referring now to FIG. 10, a system 3000 for detecting tissue and foreign objects during a surgical operation is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. 10, the system 3000 can include a surgical instrument 3002, which can be similarly configured to the surgical instrument 10000 of FIG. 1 and a computing device 3004. Although the surgical instrument 3002 of FIG. 1 is connected to the computing device 3002 via a cable, the surgical instrument 3002 can be configured for any form of wired and/or wireless communication with the computing device 3004. As used herein, the term “computing device” can include any server, personal computer, laptop, tablet, and/or mobile phone, amongst other devices communicably coupled to a display and capable of generating a user interface associated with use of the surgical instrument 3002. According to some non-limiting aspects, the computing device 3004 can be further configured to control the surgical instrument 3002 during the surgical operation. Although, according to the non-limiting aspect of FIG. 1, the surgical instrument 3002 is a handheld surgical stapler, the present disclosure contemplates other non-limiting aspects wherein the surgical instrument 3002 is robotically configured. Alternately, the surgical instrument 3002 can be an electrosurgical instrument configured to perform surgical operations via ultrasonic and/or RF energy.

[0124] In certain aspects, the computing device **3004** is a component of a surgical hub in wired and/or in wireless communication with the surgical instrument **3000**. Various suitable surgical hubs are disclosed in U.S. patent application Ser. No. 16/209,453, titled METHOD FOR CONTROLLING SMART ENERGY DEVICES, and filed Dec. 4, 2018, which is hereby incorporated by reference herein in its entirety.

[0125] In further reference to FIG. **10**, the surgical instrument **3002** can include sensors, such as one or more electrodes **3012** positioned within an end effector **3014** coupled to the distal end of a shaft **3010**. However, according to other non-limiting aspects, other sensors can be utilized to achieve a similar effect. According to some non-limiting aspects, an array of electrodes **3012** can be positioned within the end effector **3014**. For example, according to some non-limiting aspects, the array can include 16 electrodes **3012**. According to other non-limiting aspects, the electrodes **3012** can be positioned in a segmented configuration, as previously discussed. The electrodes **3012** can be electrically coupled through the inside of a hollow cavity defined by the shaft **3010** such that they can be connected to electronics positioned within a handle portion **3008** of the surgical instrument **3002**. For example, the handle portion **3008** can include processing electronics configured to process signals passed by the electrodes **3012**. Such electronics can include impedance measuring electronics, which can determine an electrical impedance of media positioned between jaws of the end effector **3014** based on the signals passed by the electrodes **3012**. Alternately and/or additionally, electronics within the handle portion **3008** can transmit signals received from the electrodes **3012** to a connected computing device **3004** for further processing.

[0126] Accordingly, during a surgical operation, such as an intraoperative stapling procedure, the surgical instrument **3002** can be used to generate real-time electrical impedance measurements based on signals received from the electrodes **3012**. These signals can be processed along with other variables, such as selected inputs, using a model and/or algorithm, which can be used to detect media within the jaws of the end effector **3014** and determine characteristics of said media, such as a location and/or a condition of the media. Accordingly, the system **3000** of FIG. **10** can be used to detect both tissue and foreign objects (e.g., NGtubes, staples, etc.) within the jaws of the end effector **3014**. The algorithmic results can be communicated to a display of a communicably coupled computing device **3004**. According to some non-limiting aspects, the system **3000** does not require a computing device **3004** and the display (e.g., an LED screen, etc.) can be integrated into the handle portion **3008** of the surgical instrument **3002**. According to other non-limiting aspects, the display can be a component of a console for a robotically assisted device. In still other non-limiting aspects, no display is required by the system **3000** and either the surgical instrument **3002** or the computing device **3004** can be configured to emit an audible notification associated with the algorithmic results. Furthermore, according to some non-limiting aspects, one or more depicted components of the system **3000** of FIG. **10** can include a clock circuit configured to measure passing time. Accordingly, the system **3000** of FIG. **3** can be configured to communicate the algorithmic results and thus, information associated with the detected media within the jaws of the end effector **3014** to an operating clinician.

[0127] As such, an algorithm, which will be described in further detail herein, can be used in conjunction with various surgical instruments, including the instrument **3000** of FIG. **10**, or the robotically controlled instrument **5100** of FIG. **3**, to detect and/or characterize various media (e.g., tissue, foreign objects, etc.) positioned within the jaws of an end effector in real time. Referring now to FIG. **11**, a user interface **3015** displayed by the computing device **3004** of the system **3000** of FIG. **10** is depicted in accordance with at least one non-limiting aspect of the present disclosure. One or more components of the system **3000** (FIG. **10**) can include a memory configured to store the algorithm. For example, according to some non-limiting aspects, the algorithm can be stored within a memory of the computing device **3004**. However, according to other non-limiting aspects, the algorithm can be stored within a memory of the surgical instrument **3002**. Regardless, the

algorithm can be implemented to generate one or more components of the user interface **3015** of FIG. **11**.

[0128] According to the non-limiting aspect of FIG. **11**, the user interface **3015** can include an image of the end effector **3014** of the surgical instrument **3002** (FIG. **10**) grabbing a tissue sample **3016** within a patient. For example, the image can be generated via one or more image sensors implemented by the system **3000** (FIG. **10**). For example, the image sensor can be mounted on or around the surgical instrument **3002** (FIG. **10**). The user interface **3015** can further include a widget **3018** overlaying the image, with a digital representation of the end effector **3020**. The digital representation of the end effector **3020** can include a first portion **3022** corresponding to the tissue sample **3016** within the jaws of the end effector **3014**, as detected by the electrodes **3012** (FIG. **10**). The digital representation of the end effector **3020** can further include a second portion **3024** corresponding to a foreign object within the jaws of the end effector **3014**, as detected by the electrodes **3012** (FIG. **10**). Accordingly, the widget **3018** can visually communicate the specific composition and/or location of various media positioned within the jaws of the end effector **3014** to an operating clinician. This can enhance the surgical operation because the operating clinician can identify exactly what objects are positioned within the end effector **3014** as well as the specific location of those objects within the end effector **3014**. In other words, the system **3000** of FIG. **10** can provide in-operative, real-time feedback regarding the location of tissue and/or foreign objects within the jaws of the end effector **14**. This feedback can be implemented to prevent inadvertent damage and/or critical errors to surrounding structures during a surgical operation, such as a stapling procedure. Detection algorithms can be light and efficient and thus, suitable for on-board electronics deployment for both handheld and/or robotics surgical platforms.

[0129] Algorithms can generate the digital representation of the end effector **3020**—including the first and second portions **3022**, **3024**—based on media parameters (e.g., impedance) that can be algorithmically determined based on signals received from the electrodes **3012** (FIG. **10**) positioned within the end effector **3014**. For example, the electrodes **3012** (FIG. **10**) can be utilized to detect an electrical impedance of the media positioned within the jaws of the end effector **3014**, which can vary significantly based on the material composition of the media. Accordingly, the determined electrical impedance can correspond to a particular type of media positioned within the jaws of the end effector **3014**, such as different types of tissue, other biological material with varying properties (e.g., blood, blood vessels, veins, etc.), an interaction between a tool and/or tissue, a foreign object (e.g., staples, tubes, etc.), and/or varying clinical environments the end effector is placed in, amongst other forms of media.

[0130] However, it may be difficult to identify the boundaries of parameter variations, such as electrical impedance, in certain situations. Accordingly, in some non-limiting aspects, an algorithm can use signals from the electrodes **3012** (FIG. **10**) to first detect non-tissue media within the jaws of the end effector **3014**, which may correspond to anomalous parameters (e.g., impedances) relative to biologic media positioned within the jaws. In other words, non-tissue media between the jaws of the end effector **3014** may be more easily classified, as they may only correspond to a few categories of media. Media such as air, liquids, and/or foreign objects positioned within the end effector **3014** can result in signals that will result in different algorithmically determined parameters (e.g., impedances) relative to biologic media (e.g., tissue) positioned within the end effector **3014** and thus, such media can be easier to detect and distinguish via a user interface, such as the user interface **3015** of FIG. **11**.

[0131] Referring now to FIG. **12**, a chart **3100** illustrating a parameter (e.g., impedance) measured by the system **3000** of FIG. **10** in time is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **12**, the chart **3100** can include illustrate how a magnitude of the impedance can vary by location, as illustrated by the x-axis, and time, as illustrated by the y-axis. The location can be determined via the impedance measured based on electrical parameters (e.g., voltage) generated by various groupings of

electrodes **3012** (FIG. **10**) installed within the end effector **3014** (FIG. **10**). Time can be measured via any desired duration, though according to some aspects time can be measured in seconds to promote the real-time capabilities of the system **3000** (FIG. **10**) and algorithm.

[0132] In further reference to FIG. **12**, the chart **3100** conceptually illustrates the detection results, including classifications of media detected within the end effector **3014** (FIG. **10**), which can be illustrated using different colors and/or patterns to enhance the visual distinction along the x-axis as time elapses, as represented along the y-axis. In other words, the chart **3000** of FIG. **12** can represent impedance detected by various groups of electrodes **3012** (FIG. **10**) positioned within the end effector **3014** (FIG. **10**) in time, which can be used to characterize the composition and location of various media positioned within the end effector **3014** (FIG. **10**) as the jaws of the end effector **3014** (FIG. **10**) open and close, clamping down on and compressing said media. As illustrated by the chart **3100** of FIG. **12**, an algorithm, can be used to detect a first media **3102** positioned between electrode group numbers 0 and 5, a second media **3104** positioned between electrode group numbers 5 and 10, a third media **3106** positioned between electrode group numbers 10 and 20, and a fourth media **3108** positioned beyond electrode group number 20. For example, the algorithm may determine, based on impedance determined based on signals received from the electrodes, that the first media **3102** is a fluid (e.g., blood), the second media **3104** is a foreign object (e.g., a staple, a NGtube, etc.), the third media **3106** is a tissue sample, and the fourth media **3108** is air, meaning the end effectors are open and/or nothing is positioned between the jaws beyond electrode group number 20. In fact, the third media **3106** is identifiable as a tissue sample based upon the compression exhibited by the stepped peak of impedance signal detected between electrode group numbers 10 and 20. It shall be appreciated that, as used herein, an impedance signal can include a variety of signal parameters, including a magnitude and/or a phase, amongst others. Moreover, impedance signals and their respective parameters can be measured in the time domain and/or the frequency domain, amongst others.

[0133] Referring now to FIG. **13**, a logic flow diagram of a method **3200** of detecting and locating media within the jaws of the end effector **3014** (FIG. **10**) of the system **3000** of FIG. **10** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to some non-limiting aspects, the system **3000** (FIG. **10**) can include a control circuit, such as the control circuit **2760** of FIG. **9**, and a memory configured to control an algorithm, wherein the algorithm is configured to cause the control circuit to perform the method **3200** of FIG. **13**. According to some non-limiting aspects, the control circuit and/or memory can be positioned within the surgical instrument **3002** (FIG. **10**). According to other non-limiting aspects, the control circuit and/or memory can be positioned within the computing device **3004** (FIG. **10**). According to still other non-limiting aspects, the memory can be positioned in a remotely located server (not shown), such as a cloud-based analytics system configured to communicate with various surgical instruments **3002** (FIG. **10**) and computing devices **3004** (FIG. **10**).

[0134] Specifically, FIG. **13** illustrates the main steps of data flow associated with the determination of a parameter (e.g., impedance, etc.), including recordation **3206**, processing **3214**, and result generation and communication **3219**. The algorithm logic will allow the detection and feedback of tissue locations and foreign objects, if present, within the jaws of the end effector **3014** (FIG. **10**), with block-level placeholders for the algorithmic detection of anomalies and foreign objects **3216** and the detection foreign objects **3218**, which are described in more detail in reference to FIGS. **14** and **15**, respectively. Of course, the method **3200** of FIG. **13** is merely illustrative, and the illustrated processes of the method **3200** are non-exclusive. According to other non-limiting aspects, the method **3200** may include more steps or fewer steps than are depicted in FIG. **13**.

[0135] According to the non-limiting aspect of FIG. **13**, the method **3200** can include commencing **3202** the detection process, including initiating and activating **3204** the sensors (e.g., electrodes **3012** of FIG. **10**) to take measurements. If the sensors **3012** (FIG. **10**) are not activated for measurements, the sensors **3012** (FIG. **10**) should be activated. The method **3200** can further

include recording **3206** data associated with the determined parameter (e.g., impedance, etc.) based on signals received from electrodes **3012** (FIG. **10**) of the surgical instrument **3002** (FIG. **10**) and checking **3208** the signal quality. If the quality exceeds a predetermined margin of error **3210**, the system **3000** (FIG. **10**) may issue a warning **3211** that the sensors have failed or are generating data of insufficient quality. According to some non-limiting aspects, the warning can be a visual alert provided via a display, such as a display coupled to the computing device **3004** (FIG. **10**) or embedded within the surgical instrument **3002** (FIG. **10**), itself. According to other non-limiting aspects, the warning can be an audible alert provided via the surgical instrument **3002** (FIG. **10**) and/or computing device **3004** (FIG. **10**).

[0136] Still referring to FIG. **13**, the method **3200** can further include filtering and preprocessing **3212** data streams associated with the surgical instrument **3002** (FIG. **10**), and processing **3214** them accordingly using an algorithm. The method **3200** can further include algorithmically detecting **3216** whether there is a foreign object **3218** and/or fluid **3220** within the jaws of the end effector **3014** (FIG. **10**). If either determination is affirmative, the system **3000** (FIG. **10**) can issue **3219** a warning to the operating clinician. Once again, the warning can either include a visual and/or audible alert. If not, the method **3200** can include determining **3222** if there is an open circuit at the location, or, determining **3224** if there is tissue at the location. The method **3200** can further include determining **3226** whether the detection methods should be repeated for a subsequent time period and, if the determination is affirmative, the method **3200** can be repeated from the filtering and preprocessing **3212** step. If the algorithm determines that the detection method should not be repeated, the method **3200** can be terminated **3228**.

[0137] In other words, the method **3200** can include sensing that is automatically triggered/turned on by users (e.g., surgeons, operating clinicians, etc.) during a procedure, such as a surgical stapling procedure. The method **3200** can include the determination of real-time parameters, such as impedance measurements, based on signals received from sensors, such as each electrode **3012** (FIG. **10**) within the end effector **3014** (FIG. **10**), which are algorithmically processed to determine an impedance characteristic (or “feature”) of the media positioned within the end effector **3014** (FIG. **10**). According to some non-limiting aspects, the impedance feature can be the impedance magnitude at a pre-determined frequency. According to some non-limiting aspects, one or more frequencies can be selected. For example, it may be preferable to use a frequency of approximately 100 kHz. Furthermore, the method **3200** can record and track impedance features over time. Once determined, such impedance features can be used as inputs to the media detection model and/or algorithm.

[0138] A detection model, via built-in algorithmic logic, can differentiate the presence and/or location of various media, such as tissue and non-tissue mediums, positioned within the jaws of the end effector **3014** (FIG. **10**), which will be described in further detail in reference to FIG. **14**. Non-tissue media can be air, indicating that the jaws of the end effector **3014** (FIG. **10**) are open. Alternately and/or additionally, non-tissue media can include liquids (e.g., blood) and/or foreign objects (e.g., staples, NGtubes, etc.). The non-tissue media can be detected via anomalies. The location of media such as tissue can then be detected after non-tissue mediums location are effectively identified. The algorithm can then classify the results, marking it as tissue, air, liquids, and/or foreign objects based on the determined impedances, and the classified results can be visually displayed via the system **3000** (FIG. **10**). Furthermore, the method **3200** can include combining detection results from pairs of electrodes **3012** (FIG. **10**) (e.g., a driven electrode and corresponding return electrode) of an array of electrodes **3012** (FIG. **10**) to provide the location information of the detected tissue and non-tissue media along the jaws of the end effector **3014** (FIG. **10**). According to some non-limiting aspects, some detection results can be selectively displayed in real-time, depending on a particular step of the surgical operation. As such, the system **3000** (FIG. **10**) can provide critical and timely feedback to operating clinicians.

[0139] Referring now to FIG. **14**, a logic flow diagram of the method **3216** of detecting an anomaly

as part of the method **3200** of FIG. **13**, is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **14**, data outputs from the processing **3214** step of the method **3200** of FIG. **13** can be used to perform the anomaly detecting **3216** step of the method **3200** of FIG. **13**, which begins by extracting, or algorithmically determining **3232**, an impedance signal value at selected frequencies for each electrode **3012** (FIG. **10**) of an electrode group positioned within the jaws of the end effector **3014** (FIG. **10**) of the system **3000** of FIG. **10**. Once again, according to some non-limiting aspects, one or more frequencies can be selected. The determination **3232** can be based on signals received from each electrode **3012** (FIG. **10**). The determined impedance signal values can be correlated with a particular time within the time domain, illustrated along the y-axis of the chart **3100** of FIG. **12**. According to other non-limiting aspects, the determined parameter can be another characteristic of the media positioned within the jaws of the end effector **3014** (FIG. **10**) and can be determined based on signals received from sensors other than the electrode **3012** (FIG. **10**).

[0140] According to the non-limiting aspect of FIG. **14**, once the impedance value is determined, the method **316** can further include a series of comparisons **3234**, **3236**, **3218** of the determined impedance value to one or more predetermined thresholds. For example, according to the non-limiting aspect of FIG. **14**, the method **3216** can include a determination **3234** of whether the determined impedance value is less than a predetermined threshold associated with an open circuit. If the determined impedance value is not less than the predetermined threshold associated with an open circuit, the algorithm can determine that open jaw (air media in jaw) has been detected, and classify **3235** the associated electrode **3012** (FIG. **10**) group as “open.” If the determined impedance value is less than the predetermined threshold associated with an open circuit, the method **3216** can further include a determination **3236** of whether the determined impedance value is greater than a predetermined threshold associated with a fluid. If the determined impedance value is not greater than the predetermined threshold associated with a fluid, the algorithm can determine that fluid has been detected, and classify **3237** the associated electrode **3012** (FIG. **10**) group as “fluid.” If the determined impedance value is greater than the predetermined threshold associated with a fluid, the method **3016** can proceed to a determination **3218** of whether a foreign object is present within the jaws of the end effector **3014** (FIG. **10**), as illustrated in the method **3200** of FIG. **13**. If the algorithm determines that no foreign object is present within the jaws of the end effector **3014** (FIG. **10**), the algorithm can determine that tissue is present within the jaws of the end effector **3014** (FIG. **10**) and classify **3239** the associated electrode **3012** (FIG. **10**) group as “tissue.” If a foreign object is detected within the jaws of the end effector **3014** (FIG. **10**), the method **3016** can proceed to a determination **3240** of whether the foreign object is detected for a predetermined time and for more than two neighboring electrode **3012** (FIG. **10**) groups. If the algorithm determines that the foreign object is detected for a predetermined time and for more than two neighboring electrode **3012** (FIG. **10**) groups, the algorithm can determine that a foreign object is present within the jaws of the end effector **3014** (FIG. **10**), classify the associated electrode **3012** (FIG. **10**) groups as “foreign object,” and record a location within the end effector **3014** (FIG. **10**) that is associated with the foreign object. In fact, the algorithmic method **3216** can further record a location within the end effector **3014** (FIG. **10**) that is associated with the open classification, the fluids classification, and/or the tissue classification, as well. If the algorithm determines that the foreign object is not detected for a predetermined time and/or that the foreign object is not detected for more than two neighboring electrode **3012** (FIG. **10**) groups, the algorithm can determine **3240** that there is noise and/or a nuisance that led to the initial detection **3218** of the foreign object.

[0141] In further reference to the non-limiting aspect of FIG. **14**, it may be preferable to calibrate **3242** the electrodes **3014** such that impedance signal values can be determined at a specific frequency or frequencies. According to some non-limiting aspects, one or more frequencies can be selected. For example, it may be preferable to determine the impedance signal values at frequencies ranging from 5 kHz to 500 kHz. According to some non-limiting aspect, it may be preferable to

determine the impedance signal values at a frequency of 100 kHz, which would allow the threshold values considered during the various comparison steps **3234**, **3236**, **3218**, **3240** can be empirically learned. Accordingly, the algorithm and/or models can be particularly configured to require fewer inputs, which can reduce a computational burden, maintain the simplicity and robustness of the method **3200**, and therefore, enable the method **3200** to be deployed via real-time, on-board electronics. Accordingly, an algorithm and methods **3200**, **3216** of FIGS. **13** and **14** represent a technological improvement over conventional methods of detecting media within the jaws of an end effector, which can require extensive processing requirements that may be prohibitive to on-board deployments. Experimental results, such as those generated in the experiment described in reference to the user interface of FIG. **18**, exhibit that the algorithmic methods **3200**, **3216** of FIGS. **13** and **14** can be more efficient and effective than conventional means of media detection, improve surgical outcomes, and can enhance patient safety relative to conventional means.

[0142] Referring now to FIG. **15**, a logic flow diagram of the method **3218** of detecting the presence of a foreign object within the jaws of an end effector **3014** (FIG. **10**), as part of the methods **3200**, **3216** of FIGS. **13** and **14**, is depicted in accordance with at least one non-limiting aspect of the present disclosure. As previously discussed, the algorithmic method **3216** of FIG. **14** can detect **3244** and classify air and/or a medium positioned within the jaws of the end effector **3014** (FIG. **10**). For example, the algorithmic method **3216** (FIG. **14**) can detect **3244** an open circuit and thus, a significant drop in the impedance value determined between one or more electrode **3012** (FIG. **10**) groups. Alternately and/or additionally, the algorithmic method **3216** (FIG. **14**) can detect **3244** a media that has contacted the jaws of the end effector **3014** (FIG. **10**) based on a magnitude of the determined impedance value. As previously discussed, the impedance signals can be determined as a function of frequency and/or multiple frequencies, electrode **3012** (FIG. **10**) group, and/or time, depending on user preference and/or intended application.

[0143] According to the non-limiting aspect of FIG. **15**, the method **3218** can further include tracking **3246** the determined impedance signals over time and record certain data points of interest. For example, it may be preferable to record a minimum impedance signal value tracked during a period of time. The method **3218** can further include calculating **3248** a ratio of any determined impedance signal value relative to the data points of interest, such as a minimum impedance signal value tracked during a period of time. The method **3218** can further include determining **3250** whether the ratio is greater than a predetermined threshold, such as a predetermined threshold over a predetermined time period. If the algorithm determines that the ratio is not greater than the predetermined threshold, it will continue calculating **3248** ratio in time. However, if the algorithm determines that the ratio is greater than the predetermined threshold, it can determine **3252** that a foreign object (e.g., a staple, a bougie, an NGtube, etc.) is present within the jaws of the end effector **3014** (FIG. **10**).

[0144] Still referring to FIG. **15**, according to some non-limiting aspects, the featured input for the algorithmic method **3218** of foreign object detection can be a calculated ratio of impedance signal at a specific frequency relative to its minimum value marked at jaw-tissue contacting moment. For example, according to some non-limiting aspects, it may be preferable to calculate the impedance signals and ratios at a frequency of 100 kHz. Additionally and/or alternately, threshold values can be empirically learned. According to the non-limiting aspect of FIG. **15**, the method **3218** can be deployed to detect foreign objects such as NGtubes, and/or other tube-like objects. However, according to other non-limiting aspect, other foreign objects can be detected using similar techniques. Experimental results of the method **3218** of FIG. **15** are further described in reference to FIG. **17**, including a distinct pattern of impedance responses for NGtube and/or a tissue sample positioned within the jaws of an end effector **3012** (FIG. **10**). It can be possible to characterize different impedance response patterns associated with different foreign objects of interest. Correspondingly, the algorithmic logic can be altered to detect various foreign objects, while employing substantially a similar functional approach, as illustrated in FIG. **15**.

[0145] Referring now to FIG. 16, a chart 3300 illustrating example results of the methods 3200, 3216, 3218 of FIGS. 13-15 is depicted in accordance with at least one non-limiting aspect of the present disclosure. As depicted by the chart 3300 of FIG. 16, different media 3304, 3306 3308, 3310, 3312 can have different determined impedance signals at different frequencies. For example, the chart 3300 of FIG. 16 illustrates impedance signals for blood 3304, lung tissue 3306, peritoneal fluid 3308, saline 3310, and stomach tissue 3312. Moreover, the chart of FIG. 16 depicts impedance signals taken over a particular frequency range of interest 3302. According to some non-limiting aspects, the frequency range of interest can be approximately between 10 kHz and 150 kHz. Furthermore, the chart 3300 of FIG. 16 depicts how a group of fluids 3314 can have lower determined impedance signals relative to other media detected within the jaws of the end effector 3014 (FIG. 10). In other words, non-tissue media, including the blood 3304, the peritoneal fluid 3308, and the saline 3310 can be easily differentiable from tissue, such as the lung tissue 3306 and the stomach tissue 3312, as previously discussed. Specifically, the chart 3300 of FIG. 16 is a bode (e.g., frequency vs magnitude) boxplot that summarizes impedance signal variation of all the data collected from various experiments for tissue and fluids, including pig lung 3306, pig stomach tissue 3312, saline 3310, blood 3304, and peritoneal fluids 3308. However, it is anticipated that similar results, including the distinguishable difference in impedance signals determined for the fluid 3304, 3308, 3310, would be present for human-oriented implementations.

[0146] Referring now to FIGS. 17A-C, another chart 3400 illustrating experimental results of the methods of FIGS. 13-15 is depicted in accordance with at least one non-limiting aspect of the present disclosure. Specifically, FIGS. 17B and 17C illustrate portion of interest of chart 3400 in more detail. Furthermore, the chart 3400 has been annotated to illustrate notable features of the experimental results. According to the non-limiting aspect of FIGS. 17A-C, the chart 3400 can illustrate various impedance signals determined based on signals received from different electrode 3012 (FIG. 10) groups as time elapses. FIG. 17C, specifically, illustrates how patterns of determined impedance responses in time can be quite different for electrode 3012 (FIG. 10) groups associated with locations where a NGtube is present, especially when compared to impedance responses associated with electrode 3012 (FIG. 10) groups associated with locations where a NGtube is not present and only tissue exists between the jaws of the end effector 3014 (FIG. 10). As time elapses along the domain, the jaws of the end effector 3014 (FIG. 10) are closing and thus, compressing media (e.g., NGtube, tissue) positioned between the jaws. As the jaws are compressing, particularly between 30 seconds and 40 seconds along the time-domain, the impedance signal is rising at a more dramatic rate for electrode groups 6-10, where a foreign object, such as an NGtube, is present. This pattern is demonstrably different than the relatively stable impedance signal determined at electrode groups 5 and 11-13, where only tissue is present. Accordingly, the system 3000 (FIG. 10) and more specifically, the control circuit, can generate an alert that communicates the presence and specific location of the detected foreign object to an operating clinician. According to some non-limiting aspects, the alert can be visually displayed via a user interface, such as the user interface 3015 of FIG. 11. According to other non-limiting aspects, the alert can be audibly communicated via a speaker positioned within the surgical instrument 3002 (FIG. 10) or the computing device 3004 (FIG. 10).

[0147] Referring now to FIG. 18, another user interface 3500 displayed by a computing device 3004 of the system of FIG. 10 is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. 18, the user interface 3500 of FIG. 18 can display image data and a visual representation of the algorithmic results generated during a simulated experiment utilizing the methods 3200, 3216, 3218 of FIGS. 13-15. Specifically, the user interface 3500 can include an image portion 3502 and a digital representation 3506 of an end effector. However, according to the non-limiting aspect of FIG. 18, test equipment is simulating the jaws of an end effector. Once again, the image can be generated via one or more image sensors implemented by the system 3000 (FIG. 10). The user interface 3500 can further include a widget

3504 overlaying the image portion **3502**, with a digital representation of the end effector **3506**. [0148] As illustrated in FIG. **18**, the simulated end effector is compressing various media. For example, a distal portion of the simulated end effector has no media beneath it, a center portion of the simulated end effector has an NGtube beneath it, and a proximal end of the simulated end effector has tissue beneath it. As previously discussed one or more electrodes **3012** (FIG. **10**) can be positioned within an end effector **3014** (FIG. **10**), or in the case of FIG. **18**, the simulated end effector. The widget **3504** and, more specifically, the digital representation of the end effector **3506** can be color coded, wherein each color can be associated with a media detected by the algorithmic methods **3200**, **3216**, **3218** of FIGS. **13-15**, based on calculated impedance signals based on signals passed by the one or more electrodes positioned within the simulated end effector. According to the non-limiting aspect of FIG. **18**, three media have been detected, including air **3508**, tissue **3510**, and a foreign object, the NGtube **3512**. The digital representation of the end effector **3506** includes a longitudinal scale, corresponding to the relative positions within which the electrodes **3012** (FIG. **10**) span from a distal end of the end effector to a proximal end of the end effector. Accordingly, the digital representation of the end effector **3506** is segmented into four colored portions, wherein each portion corresponds to media determined by the algorithmic methods, as detected by the electrodes. For example, according to the non-limiting aspect of FIG. **18**, the widget **3504** can communicate to an operating clinician that air **3508**, tissue **3510**, an NGtube **3512**, and more tissue **3510** is positioned within the jaws of the simulated end effector, from its distal to its proximal end, in order.

[0149] As will be appreciated, the capability to consistently identify the presence of a foreign object beneath the tissue being operated on is imperative. For example, foreign objects can include NGtubes, bougies, previous staple lines, and/or additional, non-tissue objects that could obstruct and/or adversely effect the surgical operation. For example, an operating clinician needs to know whether a surgical instrument, such as the surgical instrument **3002** of FIG. **10**, has accidentally clamped down upon a foreign object, prior to the firing of said surgical instrument. If a foreign object, such as an NGtube, lies beneath the tissue, it can be difficult for the operating clinician to know if the surgical instrument has actually clamped on the NGtube, which is not ideal. This lack of certainty and visibility as to what media is between the jaws of the end effector can complicate a surgical operation and lead to less desirable surgical outcomes. Accordingly, to detect and differentiate the presence of these foreign objects from the surrounding tissue, several key patterns have been observed, and an algorithm has been developed to detect the presence of foreign objects. Some of the patterns contemplated by the present disclosure will be discussed in further detail, below.

[0150] Referring now to FIG. **19**, another chart **3600** illustrating experimental results of the methods of FIGS. **13-15** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **19**, the algorithmic methods **3200** of FIGS. **13-15** have detected stomach tissue **3602** and a foreign object **3604**, such as an NGtube, between the jaws of the end effector **3014** (FIG. **10**). However, as the jaws of the end effector **3014** (FIG. **10**) close in time, the determined impedance signal associated with the NGtube **3604** increases whereas the determined impedance signal associated with the stomach tissue **3602** remains relative uniform. Accordingly, this pattern can be used by the algorithmic methods **3200**, **3216**, **3218** of FIGS. **13-15** and the system **3000** of FIG. **10** to generate a user interface, such as the user interface **3500** of FIG. **18**, which can communicate to the operating clinician that a foreign object, such as the NGtube **3604**, is present between the jaws of the end effector **3014** (FIG. **10**), as well as the precise location of the NGtube **3604** within the jaws.

[0151] Referring now to FIG. **20**, another chart **3700** illustrating experimental results of the methods of FIGS. **13-15** is depicted in accordance with at least one non-limiting aspect of the present disclosure. The chart **3700** of FIG. **20** illustrates a second identifying pattern which can be used by the algorithmic methods **3200**, **3216**, **3218** of FIGS. **13-15** to detect media positioned

between the jaws of the end effector **3014** (FIG. **10**). Specifically, the chart **3700** of FIG. **20** depicts a change in impedance phase over time, which produces a pattern that can be used to differentiate foreign objects from tissue. For example, according to the non-limiting aspect of FIG. **20**, the algorithmic methods **3200** of FIGS. **13-15** have once again determined impedance signals associated with stomach tissue **3702** and a foreign object **3704**, namely, a NGtube, between the jaws of the end effector **3014** (FIG. **10**). However, according to the non-limiting aspect of FIG. **20**, the phase of the foreign object **3704** decreases over time, while the phase for stomach tissue stays relatively constant. Accordingly, this pattern can be used by the algorithmic methods **3200**, **3216**, **3218** of FIGS. **13-15** and the system **3000** of FIG. **10** to generate a user interface, such as the user interface **3500** of FIG. **18**, which can communicate to the operating clinician that a foreign object, such as the NGtube **3704**, is present between the jaws of the end effector **3014** (FIG. **10**), as well as the precise location of the NGtube **3704** within the jaws.

[0152] According to some non-limiting aspects, a combination of patterns and trends, including the patterns illustrated in FIGS. **19** and **20**, can be collectively implemented to distinguish foreign objects from tissue. Referring now to FIG. **21**, a logic flow diagram of a method **3800** that collectively implements multiple patterns to distinguish detected foreign objects from detected tissue within the jaws of an end effector is depicted in accordance with at least one non-limiting aspect. According to the non-limiting aspect of FIG. **21**, the method **3800** can include the collection of impedance input data **3802**, feature extraction **3804**, and model development **3806**. As previously described, the method **3800** can commence via the receipt of signals from sensors, such as electrodes **3012** (FIG. **10**) positioned within the jaws of an end effector **3014** (FIG. **10**). Based on such signals, an algorithm can be configured to determine parameters such as impedance signal and impedance phase in time, as illustrated by the input step **3802** of the method **3800** of FIG. **21**. These determined measurements can be used to characterize, or extract features **3804**, such as a phase shift in time and/or a change in impedance signal in time. These features, and other data, can also be labeled as part of the feature extraction **3804**. Finally, the model can be developed **3806**. Model development **3806** can include the use of the determined phase shift or change in impedance signal to distinguish various media within the jaws of the end effector **3014** (FIG. **10**). As such, an output from the method **3800** of FIG. **21** can be used to identify various media, such as foreign objects, within the jaws of the end effector **3014** (FIG. **10**) and thus, used by the system **3000** (FIG. **10**) to provide a warning to an operating clinician.

[0153] According to other non-limiting aspects, the present disclosure contemplates a support vector machine model for classification of tissue location within the jaws of the end effector **3014** (FIG. **10**). In order to provide tissue location feedback to the surgeon, a robust algorithm that can differentiate between tissue, fluid, and air must be developed. It has been observed that this distinction can be made when using Electrical Impedance Spectroscopy (“EIS”) technology. Machine learning techniques and/or support vector machines can be used to provide such classifications. Such techniques can further assist an operating clinician in visualizing where tissue is located within the jaws of the end effector **3014** (FIG. **10**), to ensure that an entire vessel is fully clamped before performing a surgical operation, such as cutting and/or stapling. This can be challenging to visualize when an operating clinician is operating deep within the body and cannot see the cut line at the distal tip of the device. For example, when firing a surgical instrument, such as a stapler, across a vessel or artery, it may be important to capture the vessel fully within the jaws of the device. Capturing the vessel fully within the jaws can ensure that the surgical operation, such as a cutting and/or sealing of the vessel, occurs all the way across the vessel, thereby preventing blood loss. Sometimes during a procedure, however, it can be difficult to visually identify where the tissue is located within the jaws and if it extends past the cut line.

[0154] Referring now to FIG. **22**, a logic flow diagram of a method **3900** of training an algorithmic model to intelligently classify the location of media within the jaws of an end effector **3014** (FIG. **10**) is depicted in accordance with at least one non-limiting aspect. Specifically, the method **3900**

can include the use of a support vector machine model of classifying tissue located within the jaws of the end effector **3014** (FIG. **10**). According to the non-limiting aspect of FIG. **22**, the method **3900** can include the collection of impedance input data **3902**, data scaling and feature extraction **3904**, model training **3906**, and outputting the model **3908**. As previously described, the method **3900** can commence via the receipt of signals from sensors, such as electrodes **3012** (FIG. **10**) positioned within the jaws of an end effector **3014** (FIG. **10**). Based on such signals, an algorithm can be configured to determine parameters such as impedance signal and impedance phase in time, as illustrated by the input step **3902** of the method **3900** of FIG. **22**. These determined measurements can be used to for data scaling and feature extraction **3904**, such as scaling a phase shift (e.g., by $\frac{1}{2}$ rad, etc.), scaling an impedance signal (e.g., by 1/100 Kohms, etc.), determining a phase shift in time, and/or determining a change in impedance signal in time. For example, the impedance phase (“Zphase”) can be scaled between a range of 0.001π and 2π radians and the impedance magnitude (“Zmag”) can be scaled between a range of 0.001 and 1 Mohm. The scaling can be performed using a custom scaling function. These features, and other data, can also be labeled as part of the feature extraction **3904**. Calculations can be performed on the data to extract new features such as change in Z magnitude over time. However, the scaling and feature extraction **3904** can differ from a corresponding step **3804** of the method **3800** of FIG. **21** in that the data can be labeled with the appropriate tissue type and the input features with their tissue type labels are used to train **3906** a the algorithmic model. For example, the method **3900** can include training **3906** the model via a linear support vector machine algorithm using the open source Scikit Learn Python library.

The trained model produces a weight matrix and bias vector that are used for classification of future samples.

[0155] For example support vector machine models is one of the most popular supervised learning algorithms, which can be used for classification as well as regression problems. Specifically, the present disclosure contemplates implementing support vector machine modeling for classification of media within the jaw of an end effector **3014** (FIG. **10**) as a means of machine learning and thus, improving the algorithmic performance. The goal of the support vector machine algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that new data points (e.g., impedance measurements) can be properly and efficiently categorized in the future. This best decision boundary can be called a hyperplane. Thus, the model training **3906** can choose extreme points/vectors in time that help the algorithmic model create a hyperplane. These extreme cases are called as support vectors, and hence algorithm is known as a “support vector machine” algorithm. Then, the linear support vector machine can create a novel weight matrix, which can include weights configured to determine how significant each input feature is for each class. The training **3906** can further create a bias vector, which can be used to calculate classification scores. Finally, the model can be output **3908**. The output model **3908** can include weight matrix and/or a bias vector, as previously discussed. Ultimately, the training **3906** can be performed using a training dataset and, once trained, the algorithm can classify future samples using only a simple dot product and addition.

[0156] Referring now to FIG. **23**, a block diagram of a method **4000** of classifying a detected media after an algorithmic model has been trained via the method **3900** of FIG. **22** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **23**, the mathematical operations—which can be performed algorithmically—to classify a new sample after the model has been trained are illustrated. Specifically, the weight matrix will be an “m” by “n” matrix, where “m” (#rows) represents the number of classes the model is trained on, and “n” (#columns) represents the number of input features for the model, as extracted **3902** via the method **3900** of FIG. **22**. It shall be appreciated that the weight matrix can include any number of rows and/or columns as needed. The model can be trained using multiple classes and multiple inputs. According to some non-limiting aspects, there can be four classes (e.g.,

corresponding to tissue, fluid, air, and/or foreign objects) and four input features (e.g., corresponding to Z mag, Z phase, dZ/dt, and/or dPhase/dt). For example, according to the non-limiting aspect of FIG. 23, the model can be trained for two distinct classes, with three input features. Sample X of FIG. 23 illustrates feature data extracted from a new sample (e.g., an impedance measurement determined based on signals received from electrodes 3012 (FIG. 12)), which needs to be classified. The model consults the historical patterns, such as the relationship between determined impedance signal and impedance phase shifting in time for samples at a frequency of interest. Once classified, determined distinctions in classifications can easily be communicated to the operating clinician in accordance with any of the methods and/or means disclosed herein.

[0157] Referring now to FIGS. 24A and 24B, several charts 4100A, 4100B illustrating a distinction between various media detected within the jaws of an end effector 3014 (FIG. 10), as determined via the method 4000 of FIG. 23 is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. 24A, the first chart 4100A illustrates a determined distinction between detected tissue and fluids, when impedance signal is considered relative to impedance phase. Specifically, the model has generated a linear boundary line 4101 between tissue, such as stomach tissue 4106 and lung tissue 4108 and fluid, such as saline 4102 and blood 4104 respectively. In other words, the model has established clear distinction between tissue 4106, 4108 and fluids 4102, 4104.

[0158] According to the non-limiting aspect of FIG. 24B, the chart 4100B once again illustrates a determined distinction between detected tissue and fluids, when impedance signal is considered relative to impedance phase. Specifically, the model has generated a linear boundary line 4103 between tissue, such as stomach tissue 4116 and lung tissue 4118 and fluid, such as saline 4110, peri-blood 4112, blood 4114, and peritoneal fluid 4120 respectively. In other words, the model has established clear distinction between tissue 4116, 4118 and fluids 4110, 4112, 4114, 4120.

According to some non-limiting aspects, the algorithm can perform this classification and create the boundary line between datasets is a custom linear support vector machine.

[0159] Referring now to FIG. 25, another chart 4200 illustrating a distinction between various media detected within the jaws of an end effector 3014 (FIG. 10), as determined via the method 4000 of FIG. 23 is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. 25, the phase shift and impedance signal can be scaled via the scaling process 3904 of FIG. 23. For example, according to the non-limiting aspect of FIG. 25, the phase shift can be scaled by $\frac{1}{2}$ rad and the impedance signal can be scaled by 1/100 Kohms. Once again, the model has generated a boundary line 4201 that distinguishes tissue, such as stomach tissue 4204 from a foreign object, such as an NGtube 4202. However, according to the non-limiting aspect of FIG. 25, the model has also generated support vectors 4203, 4205 which can be used by the model to further determine and refine the boundary line 4201.

[0160] According to other non-limiting aspects of the present disclosure, devices, systems, and methods of data analysis to characterization tissue via electrical impedance spectroscopy are disclosed. During a surgical procedure, surgeons may encounter various situations where information about tissue characteristics and location within a stapler jaw becomes advantageous. The ability to know the type of tissue being fired upon as well as the position of the specimen in the jaw supports the surgeon's capacity to fire the stapling device effectively. Algorithm development needs feature inputs from data to form meaningful tissue insights from electrical impedance data using minimal computing power, which can promote processing efficiency and improve the computing performance of an intelligent surgical instrument. As part of the algorithm and model development process to extract tissue insights from electrical impedance spectroscopy sensing technology, feature inputs need to be established to identify key patterns within the data. Electrical impedance measurements are composed of impedance signal and impedance phase. These two values can be studied either through the lens of time or frequency. By studying the

sensed electrical impedance profiles over the time domain, different specimens may exhibit distinct and differentiable patterns which can be used to characterize both liquids and tissues. Such features represent a technical enhancement by providing continuous, real-time tissue characterization that could not be manually performed by the operating clinician and other surgical instruments and systems are incapable of. Thus, the techniques disclosed herein represent a significant technological improvement over conventional devices.

[0161] Referring now to FIGS. 26A and 2B, several charts 4300A, 4300B illustrating a distinction a means of characterizing tissue within the jaws of an end effector 3014 (FIG. 10) are depicted in accordance with at least one non-limiting aspect of the present disclosure. The charts 4300A, 4300B illustrate the use electrical impedance signal and phase shifts, as determined in the time domain via the methods and means disclosed herein, to provide innovative tissue insights. According to the non-limiting aspects of FIGS. 26A and 26B, the charts 4300A, 4300B illustrate measured parameters determined in time, as associated with a lung lobe 4302 and a bronchus 4304. Specifically, the chart 4300A of FIG. 26A illustrates an impedance profile in the time domain of lung lobe 4302 and bronchus 4304 tissue. The chart 4300B of FIG. 26B illustrates an impedance shift in the time domain of lung lobe 4302 and bronchus 4304 tissue. In reviewing both charts 4300A, 4300B, it becomes evident that the range of values between bronchus 4304 is easily differentiable from the lung lobe 4302 due to a different range of phase values uniquely exhibited by each specimen. The charts 4300A, 4300B further illustrate how measure values can change as a function of time, and how the time domain can also offer insights into tissue characterization. For example, the measured values may change as the operating clinician closes the jaws of the end effector 3104 (FIG. 10), thereby compressing the tissue and potentially changing the measured parameters, depending on the media. Additionally, by using the time domain, a clear distinction between tissue types is possible by using only one frequency of interest. The ability to use less frequencies to obtain the necessary model inputs streamlines and thus, reduces the computing power required by the sensing modality to characterize the tissue. Furthermore, limiting the frequency range may also reduce the complexity of the sensor hardware required on the surgical instrument.

[0162] Referring now to FIG. 27, a flow chart of a method 4400 of detecting media, locating media, and characterizing media positioned between the jaws of an end effector 3104 (FIG. 10) is depicted in accordance with at least one non-limiting aspect. Notably, the method 4400 of FIG. 27 summarizes the methods and functions performed by the systems and devices described above, and illustrates how powerful the system 3000 (FIG. 10) can be when implemented in a comprehensive way.

[0163] It shall be appreciated that, the foregoing aspects illustrate devices, systems, and methods that, when implemented via the aforementioned surgical instruments, can utilize various signals and signal parameters to generate tissue insights (e.g., tissue locations, foreign body notifications, critical structure notifications, tissue characterizations, etc.). Accordingly, the foregoing aspects illustrate devices, systems, and methods can enable intraoperative instrument-tissue interactions, which can allow an operating clinician to “see the unseen,” thereby augmenting surgical decision making, which can result in a safer, more efficient, and/or more precise surgical operation. However, when implemented via improved surgical instruments, the foregoing devices, systems, and methods can produce even more benefits, as will be discussed in further detail herein. For example, according to some non-limiting aspects, one or more electrodes can be electro-mechanically integrated into the jaws of an end effector, such that the surgical instrument can generate enhanced insights using a variety of methods, which shall include but not limited to electrical impedance spectroscopy (“EIS”). However, according to some non-limiting aspects, the electrodes can be implemented for alternate means of electrical sensing (e.g., monitoring voltage, current, power, impedance, or any combination of those). As such, the surgical instruments disclosed herein can be particularly configured to process motor load responses, based on

programmable conditions (e.g., end effector opening, end effector closing, tissue compression, etc.) that generate tissue responses. The surgical instruments disclosed herein can include various combinations of electrode arrays and multiplexers configured to activate various electrodes in various ways, thereby enabling the electrodes and thus, the surgical instruments to generate the previously disclosed insights. It shall be appreciated that the following aspects can be implemented via a handheld surgical instrument, such as the surgical instrument **1000** of FIG. 1, or via a robotic surgical instrument, such as the surgical instrument **5100** of FIG. 3.

[0164] In order to generate those insights, the surgical instruments disclosed herein can include specifically configured electrode arrays, which are integrated into various portions of and end effector, coupled via a particular routing configuration to a system interface located proximal to the end effector and electrode array. For example, the surgical instruments disclosed herein can include electrode arrays with a variety of electrodes of varying numbers (e.g., two, four, six, eight, etc.), geometric configurations (e.g., circular, rectangular, triangular, asymmetric, etc.), materials (e.g., gold, aluminum, titanium, stainless steel, etc.), and locations (e.g., channels, cartridge, anvil). As such, the arrays of electrodes can be used for EIS measurements, which are conveyed via electrical connections that run through the surgical instrument to a connection interface. This connects the stapler to the EIS control electronics. The routing of signals within the external boundary of the stapler could be achieved using a flexible printed circuit board or via a wireless connection. As will be disclosed herein, electrodes can be integrated within a channel, a separate consumable, a cartridge, or an anvil of the end effector, amongst other locations.

[0165] Likewise, the way signals are routed to and from the electrodes for insight generation can further improve the performance of a surgical instrument. For example, the use and/or position of a multiplexer, or switching integrated circuit, can enhance the way the surgical instrument activates and utilizes each electrode of an array. Moreover, the position of the multiplexer can enhance performance. For example, according to some non-limiting aspects, it might make sense to position the multiplexer in a channel defined by the end effector, elsewhere in the jaws of the end effector, the shaft of the surgical instrument, the handle of the surgical instrument, in a consumable or cartridge configured for placement within the end effector, or within a standalone piece of capital equipment (e.g., an electrosurgical generator). Additionally, the means by which the electrodes interface (e.g., wired or wireless) with the surgical instrument can further enhance performance. Additionally, a control circuit configured to process signals from the electrode array and generate insights can be positioned within the surgical instrument, or external to the surgical instrument, and can likewise be configured for wired or wireless interfacing. Accordingly, a portion of a cable can be dedicated to signals associated with power and data communication and can be plugged into the surgical instrument's proximal end to enable sensing as needed (e.g., plug-n-play functionality). A second portion of the cable can be connected to system capital module (e.g., a surgical hub), which can provide the power, signal generation and/or computational signal processing capabilities. According to some non-limiting aspects, the control circuit can be configured to receive signals from the electrode array via a wired or wireless interface, regardless of whether the control circuit is onboard or external to the surgical instrument. As such, various surgical instruments can include various degrees of system integration to generate the previously discussed insights as will be discussed in further detail herein.

[0166] For example, referring to FIG. 28, a surgical system **6000** is depicted in accordance with at least one aspect of the present disclosure. According to the non-limiting aspect of FIG. 28, the surgical system **6000** can include a hub **6001** configured to be communicably coupled to one or more modular surgical instruments **6008**, **6010**, which can be interchangeably connected to the hub **6001**. According to some non-limiting aspects, the hub **6001** can be similar to those disclosed in U.S. patent application Ser. No. 16/562,172 titled PORT PRESENCE DETECTION SYSTEM FOR MODULAR ENERGY SYSTEM, filed on Sep. 5, 2019 and published as U.S. Patent Application Publication No. 2020/0078113 on Mar. 12, 2020, the disclosure of which is hereby

incorporated by referenced in its entirety. Although the surgical instruments **6008**, **6010** of FIG. **28** are wired to the hub **6001**, according to some non-limiting aspects, the surgical instruments **6008**, **6010** can be wirelessly coupled to the hub **6001**. Regardless, the hub **6001** can communicate signals and/or commands to the surgical instruments **6008**, **6010** and can receive signals and/or feedback from the surgical instruments **6008**, **6010**.

[0167] In further reference to FIG. **28**, the surgical system **6000** can be compatibly configured for use with a robotic platform **6002** that utilizes a robotic surgical instrument **6008** and/or a handheld platform **6004** that utilizes a handheld surgical instrument **6010**, depending on the intended application. For example, according to the non-limiting aspect of FIG. **28**, the handheld surgical instrument **6010** can be a laparoscopic device; however, according to other non-limiting aspects, the present disclosure contemplates other handheld surgical instruments. In both platforms **6002**, **6004**, the hub **6001** can be positioned within a rack **6016**, **6018** that includes various control units (e.g., processors, actuators, generators, etc.) configured to control the communicably coupled surgical instruments **6008**, **6010**. For example, the robotic platform **6002**, in particular, can further include a controller of a robotic surgical system, such as the controller **5000** of FIG. **2**.

[0168] Notably, the modular surgical instruments **6008**, **6010** of FIG. **28** can each include an end effector **6012**, **6014** configured with a first jaw that is movable relative to a second jaw, as previously disclosed. Depending on the implementing surgical instrument **6008**, **6010** and/or platform **6002**, **6004**, each end effector **6012**, **6014** can be particularly configured to perform an intended surgical operation. For example, according to some non-limiting aspects, the end effectors **6012**, **6014** can be configured to perform a surgical stapling operation. According to other non-limiting aspects, the end effectors **6012**, **6014** can be configured to deliver RF and/or ultrasonic energy in response to a command received from the hub **6001** and/or energy received from a generator positioned in the rack **6016**, **6018**. For example, according to some non-limiting aspects, the end effector **6002** can be configured for an advanced bipolar surgical operation. According to still other non-limiting aspects, the end effector **6002** can be configured for Highly Articulating Tools (“HAT”), such as end effectors configured to articulate in two or more planes. Regardless, each end effector **6012**, **6014** can include an array of electrodes **6013**, **6015** configured to facilitates the generation of enhanced tissue insights using a variety of methods, as will be explained in further detail herein.

[0169] Referring now to FIGS. **29A-C**, a robotic surgical instrument **6008** configured for use with the system **6000** of FIG. **28** is depicted in accordance with at least one aspect of the present disclosure. According to the non-limiting aspect of FIG. **29A**, the surgical instrument **6008** can include an end effector **6012** configured to perform a surgical operation. For example, according to some non-limiting aspects, the end effector **6012** can be configured for use a surgical stapling operation. According to other non-limiting aspects, the end effector **6012** can be configured to deliver RF and/or ultrasonic energy. For example, according to some non-limiting aspects, the end effector **6012** can be configured for an advanced bipolar surgical operation. According to still other non-limiting aspects, the end effector **6012** can be configured for HAT.

[0170] In reference to FIGS. **29B** and **29C**, certain end effectors **6012.sub.a**, **6012.sub.b** and electrode configurations **6013.sub.a**, **6013.sub.b** that can be coupled to the robotic surgical instrument **6008** of FIG. **29A** are depicted in accordance with at least two non-limiting aspects of the present disclosure. For example, FIG. **29B** illustrates an end effector **6012.sub.a** that can include electrodes that traverse a longitudinal axis defined by the end effector **6012.sub.a** and protrude from an external wall of the end effector **6012.sub.a**. According to FIG. **29C**, electrodes can be defined from a pad that similarly traverses a longitudinal axis defined by the end effector **6012.sub.b**, but are flush relative to an external wall of the end effector **6012.sub.a**. As will be discussed in further detail, the electrodes **6013.sub.a**, **6013.sub.b** can be integrated within the end effectors **6012.sub.a**, **6012**; and/or integrated within cartridges configured to be inserted into the end effectors **6012.sub.a**, **6012.sub.b**.

[0171] In further reference to FIGS. **29A-C**, the robotic surgical instrument **6008** can be further configured to generate data associated with the robotic surgical instrument **6008**, which can enhance the feedback received from the surgical instrument **6008** and thus, improve the insights generated by a surgical hub and/or control circuit communicably coupled to the surgical instrument **6008**, as illustrated and discussed in reference to FIGS. **1-27**. For example, the surgical instrument **6008** of FIGS. **29A-C** can generate data associated with its position, load, and/or motor via one or more internal sensors (e.g., an encoder, torque sensors, controlled motion sensors, etc.) positioned within its housing. This data, in conjunction with electrical impedance spectroscopy (“EIS”) data generated by a plurality of electrodes **6013** positioned within the end effector **6012** can be used to inputs to the previously discussed algorithm engines, which can estimate, detect, and/or evaluate key outputs and/or insights regarding the tissue being operated on, the surgical instrument **6008** itself, and/or the operation, generally. Such outputs can provide high clinical values. For example, various combinations of tissue EIS data, position data, load data, and/or motor torque data generated by the surgical instrument **6008** of FIG. **29A-C** can be used by the previously discussed algorithm engines to estimate a relative tissue tension, a thickness of the tissue within the jaw, tissue characteristics, and/or the position of a critical structure (e.g., a vessel).

[0172] Referring now to FIG. **30**, a block diagram **7000** of an algorithmic engine **7002** employed by the surgical instrument **6008** of FIGS. **29A-C** is depicted in accordance with at least one non-limiting aspect of the present disclosure. Although the algorithmic engine **7002** of FIG. **30** is discussed in specific reference to the surgical instrument **6008** of FIGS. **29A-C**, it shall be appreciated that the algorithmic engine **7002** of FIG. **3** can be similarly implemented by any of the surgical instruments and/or surgical hubs disclosed herein. According to the non-limiting aspect of FIG. **30**, the surgical instrument **6008** can generate several signals **7004**, **7006**, **7008** that can be provided to the algorithmic engine **7002** for processing and/or analysis.

[0173] For example, according to some non-limiting aspects, the surgical instrument **6008** of FIGS. **29A-C** can generate signals associated with tissue EIS data **7004**, position data **7006**, and/or torque/load data **7008** detected by the various sensors and/or electrodes **6013** positioned within the end effector **6012** of the surgical instrument **6008**. The algorithm engine **7002**, employing the previously discussed methods and techniques, can process these inputs and generate one or more outputs **7010**, **7012**, **7014**, **7016** that could provide the clinician with insights regarding the tissue, surgical instrument **6008**, and/or operation being performed.

[0174] Referring now to FIG. **31**, a handheld surgical instrument **6010** configured for use with the system **6000** of FIG. **28** is depicted in accordance with at least one non-limiting aspect of the present disclosure. Although it is specifically configured for manual operation, the handheld instrument **6010** of FIG. **31** can include any of the features and provide similar benefits to those discussed in reference to the robotic surgical instrument **6008** of FIGS. **29A-C**. According to the non-limiting aspect of FIG. **31**, the surgical instrument **6010** can include an end effector **6015** and a handle **6020**. The surgical instrument **6010** can further include a shaft **6016** with a proximal portion **6018** and a distal portion **6019**. The end effector **6014** can be coupled to the distal portion **6019** of the shaft **6016** via an articulation joint and the shaft **6016** can be coupled to the handle **6020** via a nozzle **6017**. As previously described, the end effector **6014** can include a first jaw movable relative to a second jaw, and can be configured to accommodate a plurality of electrodes **6015**.

[0175] Similar to the robotic surgical instrument **6008** of FIGS. **29A-C**, the surgical instrument **6010** of FIG. **31** can also be configured to perform a surgical operation. For example, according to some non-limiting aspects, the end effector **6014** can be configured for use a surgical stapling operation. According to other non-limiting aspects, the end effector **6014** can be configured to deliver RF and/or ultrasonic energy. For example, according to some non-limiting aspects, the end effector **6014** can be configured for an advanced bipolar surgical operation. According to still other non-limiting aspects, the end effector **6014** can be configured for HAT.

[0176] Referring now to FIG. **32**, an exemplary electrode array **6015** is depicted in accordance with

at least one non-limiting aspect of the present disclosure. Although the electrode array **6015** of FIG. **32** is depicted as a component of the end effector **6014** of FIG. **31**, it shall be appreciated that the electrode array **6015** of FIG. **32** can be implemented via any of the end effectors employed by any of the surgical instruments disclosed herein. For example, a first jaw of the end effector **6014** of FIG. **31** can define an elongate channel **6021** that traverses a longitudinal axis L of the end effector **6014** (FIG. **31**). Specifically, the channel **6021** can be defined by one or more walls **6025.sub.a**, **6025.sub.b** that extend along the longitudinal axis L on opposing sides of the longitudinal axis L. For example, the electrode array **6015** of FIG. **32** includes eight pairs of electrodes **6022**, **6024**, each having a rectangular shape and constructed from titanium. However, according to other non-limiting aspects, the array **6015** can include electrodes **6022**, **6024** of varying numbers, geometries, and/or materials, depending on intended application and/or user preference. As further illustrated by FIG. **32**, certain electrodes **6024** of the array **6015** can be configured different relative to other electrodes **6022**. For example, certain electrodes **6024** can be positioned about a cutline **6026** can be configured such that the electrodes **6024** provide an increased resolution at either side of the cutline **6026**.

[0177] According to the non-limiting aspect of FIG. **32**, the electrode array **6015** can include one or more electrodes **6022**, **6024** integrated into the walls **6025.sub.a**, **6025.sub.b** of the channel **6021** defined by one of the jaws of the end effector **6014** (FIG. **31**). Specifically, each electrode **6022**, **6024** can be over-mounted, or over-molded onto the walls **6025.sub.a**, **6025.sub.b** of the channel **6021**. Of course, other means of integration can be employed to achieve a similar effect.

According to the non-limiting aspect of FIG. **32**, certain electrodes **6024** can be positioned about a cutline **6026** of the end effector **6014** (FIG. **31**), such that those electrodes **6024**, when activated by a multiplexed signals conveyed via a multiplexer (as will be discussed in further detail herein), can cut tissue about the cutline **6026**. The multiplexed signals and multiplexer can utilize the other electrodes **6022** to generate tissue insights, as previously discussed.

[0178] Referring now to FIGS. **33A-C**, another end effector **6100** configured for use with the surgical instruments disclosed herein is depicted in accordance with at least one non-limiting aspect of the present disclosure. Similar to the end effector **6014** of FIGS. **31** and **32**, the end effector **6100** of FIGS. **33A-C** can include a first jaw **6101** that defines a channel **6102** interspersed between two walls **6103.sub.a**, **6103.sub.b** of the jaw **6101**. Additionally, an array of electrodes **6104**, **6106** can be integrated within each sidewall **6103.sub.a**, **6103.sub.b** of the jaw **6101**. In specific reference to FIG. **33B**, a sectioned, isometric view of the end effector **6100** of FIG. **33A** is depicted in accordance with at least one non-limiting aspect. According to the non-limiting aspect of FIG. **33A**, the end effector **6100** can further include a flexible conductor **6108** capable of carrying multiplexed signals, wherein the flexible conductor **6108** traverses through the jaw **6101**.

[0179] In further reference to FIG. **33B**, the flexible conductor **6108** can be multiplexed and electrically coupled to the electrodes **6104**, **6106** such that the electrodes **6104**, **6106** can receive signals from transmitted by and/or through a control circuit positioned within the surgical instrument and/or surgical hub. The flexible conductor **6108** can accomplish this via multiple layers that provide electrical connections to the various electrodes **6104**, **6106**. Accordingly, the flexible conductor **6108** can be configured as a multiplexer and/or a multiple output generator, as described further below. As such, independent signals can be transmitted to and through each electrode **6104**, **6106**, such that each electrode **6104**, **6106** can serve a different purpose and/or provide a separate function relative to other electrodes **6104**, **6106** in the array. For example, tissue under each electrode **6104**, **6106** can be treated individually according to the coagulation needs. However, relevant to the generation of surgical insights, as previously discussed, certain electrodes **6104** of the array can receive signals through the flexible conductor **6108** to generate surgical insights. Other electrodes, such as the electrodes **6106** disposed about a cutline **6026** (FIG. **32**), can receive signals through the flexible conductor **6108** to cut tissue that is positioned about the cutline **6026** (FIG. **32**). Each electrode **6104**, **6106** in the array can be communicably coupled to a surgical hub

and/or a control circuit and thus, communicably coupled to an energy source, such that the electrodes **6104**, **6106** can receive energy via the flexible conductor **6108**. For example, according to one non-limiting aspect, the pairs of active electrodes **6104**, **6106** on opposite sides of the channel **6102**, may be energized by the energy source at the same time.

[0180] According to other non-limiting aspects, different pairs of segmented electrodes **6104**, **6106** can be energized or receive different signals via the flexible conductor **6108**. Various electrode pairs **6104**, **6106** can be energized by the energy source (or generator) to perform certain surgical operations and other electrode pairs **6104**, **6106** can be used to generate tissue insights, using the previously discussed techniques. The flexible conductor **6108** can convey various multiplexed signals and distribute them to the corresponding electrodes **6104**, **6106** as desired, under the control of the control circuit, which will be discussed in further detail. According to some non-limiting aspects, the energy source, the multiplexer, and the control circuit can be positioned in the nozzle **6017**, the shaft **6016**, the handle **6020**, a housing of a robotic surgical instrument, and/or within a communicably coupled surgical hub of the surgical system. According to some non-limiting aspects, the multiplexer can be similar to those discussed in U.S. patent application Ser. No. 13/795,205, titled MOTORIZED SURGICAL CUTTING AND FASTENING INSTRUMENT HAVING HANDLE BASED POWER SOURCE, and filed Feb. 14, 2008, which issued as U.S. Pat. No. 9,522,029 on Dec. 20, 2016, the disclosure of which is hereby incorporated by reference in its entirety.

[0181] According to the non-limiting aspect of FIG. **33C**, a means of integrating the electrodes **6104**, **6106** into the end effector **6100** is depicted in accordance with at least one non-limiting aspect of the present disclosure. The flexible conductor can be coupled to conductive tracks **3108.sub.a**, **3108.sub.b** that traverse each side of the end effector **3100**. Similar to the flexible conductor **3108**, each conductive track **3108.sub.a**, **3108.sub.b** can be configured to transmit multiplexed signals to and from the electrodes **3104**, **3106**. An electrode material **6110**, from which the electrodes **6104**, **6106** can be formed, can be overmolded with a material from which the sidewalls of **6103.sub.a**, **6103.sub.b** (FIG. **33A**) can be constructed. The overmolding, for example, can include a multi-step injection molding process where two or more components can be molded over top of one another, thereby resulting in a single, solid piece. Of course, according to other non-limiting aspects, other means can be employed to integrate the electrodes **6104**, **6106** into the conductive tracks **6108.sub.a**, **6108.sub.b** and sidewalls of the end effector **6100** (e.g., machining, masking, insert molding, insertion into cavities and adhesion, etc.).

[0182] The non-limiting aspect of FIGS. **33A-C** illustrate merely one of the aforementioned integration techniques. According to the non-limiting aspect of FIGS. **33A-C**, the array of electrodes **6104**, **6106** can be integrated into the channel **6102**, the flexible conductor **6108** can be configured such that multiplexing takes place within the end effector **6100**, and signals, including data signals and power signals, can be routed through the end effector **6100** via tracks on the flexible conductor **6108**. The integrated techniques employed by the end effector **6100** enable signals to be transmitted through the flexible conductor **6108**, to and from the array of electrodes **6104**, **6106**, and through an articulation joint of the surgical instrument, such as the articulation joint **6112** of FIG. **34A**. According to the non-limiting aspect of FIG. **34A**, the articulation joint **6112** can have one or more supports **6114** that a flexible conductor, such as the flexible conductor **6108** of FIGS. **33A-C**, can be routed through, thereby facilitating articulation of an end effector, such as the end effector **6100** of FIGS. **33A-C**, without interrupting any signals traversing the flexible conductor to and from the end effector. Of course, similar articulation joints **6112** can be implemented on any of the surgical instruments disclosed herein. FIG. **34B** depicts how the flexible conductor **6108** can be further proximally routed after the articulation joint **6112**, through a shaft **6116** of the surgical instrument and into a nozzle **6117**. FIG. **34C** depicts the nozzle **6117** of the surgical instrument in further detail, with the flexible conductor **6108** traversing a housing of the nozzle **6117** and electrically coupled to a control circuit **6119** positioned within the nozzle **6117**.

However, according to other non-limiting aspects, the control circuit **6119** can be positioned proximal the nozzle **6117**, or within a communicably coupled surgical hub. In some non-limiting aspects, the surgical hub can be remotely positioned relative to the surgical instrument itself. Accordingly, end effectors of surgical instruments—and more specifically, arrays of electrodes—can be effectively coupled to a control circuit proximally positioned in the surgical instrument or a communicably coupled surgical hub. The control circuit can subsequently generate insights (e.g., EIS measurements) using the signals via algorithmic implementation, as previously discussed.

[0183] Referring now to FIG. **35**, another end effector **6200** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **35**, the end effector **6200** can include one or more electrodes **6204** mounted on a separate consumable **6206** configured to be inserted within a channel **6202** defined by the sidewalls **6203.sub.a**, **6203.sub.b** of the end effector. As such, the channel **6202** and sidewalls **6203.sub.a**, **6203.sub.b** can be configured to accommodate the separate consumable **6206**. For example, the sidewalls **6203.sub.a**, **6203.sub.b** can include an interior surface composed of a conductive material, such that the conductive material is placed into electrical communication with the electrodes **6204** when the separate consumable is inserted into the channel **6202**. Alternately, the channel can include one or more electrical contacts configured to place the electrodes **6204** into electrical communication with a flexible conductor **6208** capable of carrying multiplexed signals, wherein the flexible conductor **6108** traverses through the channel **6202**.

[0184] In other words, according to the non-limiting aspect of FIG. **35**, array of electrodes **3204** can be attached to a separate consumable **6206** and thus, selectively clipped into the channel **6202**. In this way, a single end effector **6200** can be configured to accommodate numerous separate consumables **6206**, wherein each separate consumable **6206** can include a different array of electrodes **6204** of varying configurations. Furthermore, the separate consumable **6206** can be define a second channel **6212** configured to accommodate a cartridge for the surgical operation (e.g., a staple cartridge, an electrosurgical cartridge, etc.). Accordingly, various combinations of separate consumables **6206** and cartridges can be used by the same end effector **6200**. Moreover, the electrodes **6204** can receive and send signals via the flexible conductor **6208**, which can be used to generate insights per the previously disclosed techniques, regardless of the cartridge type loaded into the end effector **6200**. According to some non-limiting aspects, the flexible conductor **6208** can be routed through the end effector **6200** and surgical instrument in a method similar to those described in reference to the end effector **6100** of FIGS. **33A-C** and **34**.

[0185] Referring now to FIG. **36A-D**, another end effector **6300** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **36**, the end effector **6300** can accommodate a cartridge **6306** configured to perform a surgical operation (e.g., a staple cartridge, an electrosurgical cartridge, etc.), and an array of electrodes **6304** can be dispositioned on the cartridge **6306**, itself. In specific reference to FIG. **36B**, the sidewalls **6303.sub.a**, **6303.sub.b** of the end effector **6300** can once again define a channel **6302**, and the channel **6302** can be configured to accommodate the cartridge **6306**. The end effector can further include a flexible conductor, such as the flexible conductors **6108**, **6208** of FIGS. **33A-C**, **34** and **35**, capable of carrying multiplexed signals, wherein the flexible conductor **6308** traverses through the channel **6302**. The flexible conductor **6308** can be routed through the end effector **6300** and surgical instrument in a method similar to those described in reference to the end effector **6100** of FIGS. **33A-C** and **34**.

[0186] For example, the sidewalls **6303.sub.a**, **6303.sub.b** can include an interior surface composed of a conductive material, such that the conductive material is placed into electrical communication with the electrodes **6304** when the separate consumable is inserted into the channel **6302**, as illustrated in FIG. **36C**. Alternately, the channel can include one or more electrical contacts configured to place the electrodes **6304** into electrical communication with a flexible conductor **6308** capable of carrying multiplexed signals, wherein the flexible conductor **6308** traverses

through the channel **6302**.

[0187] Referring now to FIG. **36D**, an insertion of the cartridge **6306** into the end effector **3600** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **36D**, the end effector **6300** can include a conductive element **6312** configured to electrically interface with a corresponding conductive element **6310** on the cartridge **6300**. The conductive elements **6310**, **6312** can further be configured for multiplexed signal transmission, such that multiplexed signals transmitted through the flexible conductor **6308** of the end effector **6300** can be communicated to and from each electrode **6304** of the array, positioned on the cartridge **6306**. According to the non-limiting aspect of FIGS. **36A-D**, the array of electrodes **6304** can be integrated onto cartridge **3606**, which can be a consumable. The electrodes **6304** can be electrically integrated within the cartridge **3606** via a multiplexing integrated circuit inside the cartridge and electrical connections between the conductive elements **6310**, **6312**. Because the cartridge **6306** can include the multiplexing electronics, the interfaces **6310**, **6312** between the end effector **6300** and the cartridge **6306** can be simplified.

[0188] Referring now to FIGS. **37A** and **37B**, other end effectors **6400.sub.a**, **6400.sub.b**, are depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspects of FIG. **37A** the end effector **6400.sub.a** can be configured to accommodate a hybrid cartridge **6406.sub.a** wherein an array of electrodes **6404.sub.a** is positioned on the cartridge **6406.sub.a**. Each electrode **6404.sub.a** can be configured for electrical communication with a flexible conductor **6408** capable of carrying multiplexed signals, when the cartridge **6406.sub.a** is installed within the end effector **6400.sub.a**, wherein the flexible conductor **6308** traverses through a channel defined by the end effector **6400.sub.a** and can be routed through the end effector **6400** and surgical instrument in a method similar to those described in reference to the end effector **6100** of FIGS. **33A-C** and **34**.

[0189] In reference to FIG. **37B**, a similar, albeit subtly different, end effector **6400.sub.b** is depicted in accordance with at least one non-limiting aspect of the present disclosure. Notably, the array of electrodes **6404.sub.b** is different than the electrodes **6404.sub.a** of FIG. **37A**. As such, a plurality of conductive elements **6412** that correspond to each electrode **6404.sub.b** are dispositioned in each wall **6403.sub.a**, **6403.sub.b** of the end effector **6400.sub.b**. Accordingly, each electrode **6404**; of the array can receive an intended signal from the multiplexed signals traversing the flexible conductor **6408**.

[0190] It shall be appreciated that, according to the non-limiting aspects of FIGS. **37A** and **37B**, the array of electrodes **3404.sub.a**, **3404.sub.b** can be positioned on the cartridge **3406.sub.a**, **3406.sub.b**, which can be electrically configured with one or more electrical surfaces (e.g., metal plating, metal pieces bent around the sides, vias through the cartridge, etc.) for an intended connection of each electrode **3404.sub.a**, **3404.sub.b** to an appropriate portion of the flexible conductor **3408**. Accordingly, multiplexing takes place in the end effector **3400** but each electrode **3404.sub.a**, **3404.sub.b** still receives the appropriate signal via the electrical connections.

[0191] Referring now to FIG. **38**, another end effector **3500** is depicted in accordance with at least one non-limiting aspect of the present disclosure. Similar to the end effectors **3400.sub.a**, **3400.sub.b** of FIGS. **37A** and **37B**, the array of electrodes **3504** can be integrated on the cartridge **3506**, itself. Multiplexing, however, can occur either within the cartridge **3506** or end effector **3500** via a flexible conductor **6508** capable of carrying multiplexed signals. Regardless, the end effector **6500** of FIG. **38** can further include a wireless communication module **6514** configured to transmit and receive multiplexed signals to and from a control circuit and/or a surgical hub wirelessly via an infrastructure network (e.g., WiFi®, cellular, etc.) or an ad hoc network (e.g., Bluetooth®, Near Field Communications, RFID, etc.). Accordingly, the wireless communication module **6514** can serve as a communication interface between the end effector **6500** and the surgical hub and/or control circuit, thereby eliminating the need for the routing described in reference to FIGS. **33A-C** and **34**. It shall be appreciated that a wireless communication module **6514** of FIG. **38** can be

similarly applied to any of the surgical instruments and/or end effectors disclosed herein, thereby simplifying—and in some aspects, eliminating—the routing of the flexible conductors disclosed herein.

[0192] Referring now to FIGS. **39A-D**, another end effector **6600** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIGS. **39A-D**, the end effector **6600** can include a first jaw and a second jaw. For example, the second jaw can be configured as an anvil of the end effector **6600** and a separate consumable **6606** can be configured to be selectively coupled to the second jaw. Whereas the separate consumable **6206** (FIG. **35**) was shown as coupled to a bottom jaw of the end effector **6200** of FIG. **35**, the separate consumable **6606** can be coupled to the second, or top, jaw **6602** of the end effector **6600** of FIG. **39A**. Nonetheless, the array of electrodes **6604** can be coupled to the separate consumable **6606** of FIG. **39A** and, when coupled to conductive elements on the second jaw **6602**, electrically coupled to a flexible conductor **6608**. The flexible conductor **6608** can be capable of carrying multiplexed signals, wherein the flexible conductor **6608** traverses through the end effector **6600** and surgical instrument in a method similar to those described in reference to the end effector **6100** of FIGS. **33A-C** and **34**. Alternately, the wireless embodiment of FIG. **38** can be employed to transmit signals to and from the array of electrodes **6604**.

[0193] In reference to FIG. **39C**, a method of coupling the separate consumable **6606** to the second jaw **6602** of the end effector **6600** of FIGS. **39A** and **39B** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **39C**, an adhesive layer **6610** can be deposited on a top surface of the second jaw **6602** and the separate consumable **6606** can be deposited on top of the adhesive layer **6610**. As such, the separate consumable **6606**, including its electrodes **6604** can be coupled to the second jaw **6602**, as depicted in the cross-sectioned view of FIG. **39D**. Accordingly, the electrodes **6604** can be positioned on the underside of a printed circuit board of the separate consumable **6606**, such that they can generate signals for Of course, according to other non-limiting aspects, the separate consumable **6606** can be alternately coupled to the second jaw **6602**, for example, via corresponding geometric components, clips, fasteners, and/or a pressure fit.

[0194] Referring now to FIG. **40**, a flexible circuit **6700** configured for use with any of the end effectors disclosed herein is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **40**, the flexible circuit **6700** can be a multiple switching integrated circuit (e.g., a multiplexer) that includes one or more switch circuits **6702** positioned on a flexible printed circuit board **6704** for signal switching and control to and from an end effector. The flexible circuit **6700** can be routed through a shaft of the surgical instrument and integrated into the end effector to provide signal selection and control of electrodes at the end-effector. The flexible circuit **6700** can include a portion that is extended through the articulating joint to the end-effector where electrodes are connected. Another side of the flexible circuit **6700** can extend through the shaft to the proximal side of the tool, where signal conditioning and processing electronics can be placed. The switches **6702** can be configured to turn on/off certain electrodes, thereby resulting in different electrode configurations being activated during impedance measurements.

[0195] Referring now to FIG. **41**, a system diagram of a system **6800** configured to use the surgical instruments and end effectors disclosed herein is depicted in accordance with at least one non-limiting aspect of the present disclosure. according to the non-limiting aspect of FIG. **41**, the system **6800** can utilize a surgical instrument **6801**, that includes electronics **6802** for switching and buffering signals to and from a control circuit positioned within the surgical instrument **6801**. The surgical instrument **6801** can further include an end effector **6804** configured for a surgical operation (e.g., stapling, sealing, cutting, etc.). An electrode array **6806** can be positioned within the end effector **6804** according to any of the aspects disclosed herein. For example, the electrode array **6806** can include a specifically configured number and arrangement of electrodes (e.g., eight

pairs of electrodes, sixteen electrodes in total, etc.). The surgical instrument **6801** can further include electrical connections **6812** routed through the inside of an articulation joint and shaft of the surgical instrument **6801** using any of the flexible conductors and arrangements disclosed herein. A nozzle **6808** of the surgical instrument **6801** can include a control circuit. The nozzle **6808** can be stationary, or further configured to rotate the shaft. The surgical instrument can be further connected to a lab system **6814** or surgical hub configured to display a graphical user interface, which can include a plotting window configured to present the algorithmic insights, as previously described. The system **6800** can be compatible with a plurality of surgical instruments **6816**, each configured for use with a plurality of separate consumables and/or cartridges **6818**, as disclosed herein. Accordingly, the system **6800** can facilitate the use of numerous, customizable electrode configurations, each capable of performing different surgical operations and generating different insights.

[0196] Referring now to FIG. **42**, a surgical instrument **6900** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **42**, the surgical instrument **6900** can include a first cable **6902** input and a second cable input **6904** that enter the nozzle **6906** of the surgical instrument **6900**. As such, a control circuit **6908** positioned within the nozzle **6906** can process inputs from both the first cable **6902** and the second cable **6904** into a multiplexed signal that can traverse a flexible conductor routed through the surgical instrument **6900** using any of the methods disclosed herein.

[0197] Referring now to FIG. **43**, a diagram illustrating several non-limiting surgical system configurations **7002**, **7004**, **7006** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **43**, each of the configurations **7002**, **7004**, **7006** can employ various electrode configurations, switching circuitry, signal conditioning circuits, signal generators, signal processing circuits, and/or graphical user interfaces to generate and display the insights disclosed herein. For example, a first configuration **7002** employs an electrode array in the end effector, switching circuitry in the tool shaft, and signal conditioning in the tool stage. The signal generator and processor can be located in a surgical hub, and results can be displayed via a graphical user interface on a separate unit. According to the second configuration **7004**, an electrode array can be included in the end effector, and switching circuitry as well as signal conditioning can occur in the tool shaft. Once again, the signal generator and processor can be located in a surgical hub, and results can be displayed via a graphical user interface on a separate unit. According to the second configuration **7004**, an electrode array can be included in the end effector, and switching circuitry as well as signal conditioning can occur in the tool shaft. Once again, the signal generator and processor can be located in a surgical hub, and results can be displayed via a graphical user interface on a separate unit. According to the third configuration **7006**, an electrode array can be included in the end effector, switching circuitry can occur in the tool shaft, and as signal conditioning can occur in the tool stage. However, the signal generator and processor can be located in a surgical instrument, and results can be displayed via a graphical user interface on a separate unit, thereby bypassing the surgical hub altogether.

[0198] As previously described, the use of one or more electrodes to generate enhanced insights (e.g., tissue locations, foreign body notifications, critical structure notifications, tissue characterizations, etc.) using a variety of methods, which shall include but shall not be limited to EIS. However, according to some non-limiting aspects, similar devices, systems, and methods can be employed to expand those insights to include the identification of a particular contact and/or a particular timing of contact between the jaws of an end effector and a particular media positioned within those jaws. In other words, certain insights can be used to identify a particular contact of interest between the jaws of an end effector and a media positioned within the jaws of the end effector, which can be used as a point of reference from which other measurements can be taken. For example, such aspects can include the detection of an initial contact with a tissue, which can be subsequently used as a timing mechanism to trigger other sensors that take tissue thickness and/or

jaw displacement measurements, as will be disclosed in further detail herein.

[0199] Such applications of the aforementioned devices, systems, and methods can be particularly useful as sensing technologies are developed to measure tissue thickness and/or jaw displacement on surgical instruments. Although useful, these enhancements can make it difficult to discern exactly when the required measurements should be taken, as it can be difficult to visually identify when initial contact is made with a tissue sample. In other words, it can be difficult to determine when a sensor begins contacting the tissue while it is simultaneously sensing tissue thickness and/or end effector displacement. These problems can be addressed using the aforementioned devices, systems, and methods in accordance with the following non-limiting aspects of the present disclosure.

[0200] Referring now to FIG. **44**, a logic flow diagram of a method **4400** of identifying a particular contact between the jaws of an end effector and a media positioned within the jaws of the end effector is depicted in accordance with at least one non-limiting aspect of the present disclosure. Generally, the method **4400** of FIG. **44** utilizes the aforementioned sensing techniques (e.g., electrical impedance sensing techniques, etc.) to enable a sensor—such as a tissue-thickness detecting sensor—to be used as a timing mechanism by which the surgical instrument, or a system communicably coupled to the surgical instrument, knows when to begin taking measurements. Previously, the aforementioned sensing techniques were discussed in reference to detecting a particular media's location within the jaws of an end effector. However, according to the non-limiting aspect of FIG. **44**, similar techniques are employed to establish a particular time of contact, such as the identification of when an initial tissue contact is made. Accordingly, the method **4400** of FIG. **44** can enable baseline measurements to be taken immediately when tissue contact is initiated, such that changes in one or more sensed parameters can be tracked over time from the initial point of contact.

[0201] For example, according to the non-limiting aspect of FIG. **44**, the method **4400** can be specifically employed to perform RF impedance-based sensing to identify an initial point of tissue contact. The identified point of tissue contact can be used as a point of reference for the purposes of timing when other measurements should be taken. According to the non-limiting aspect, the method **4400** can include detecting **4402** a particular contact between the jaws of an end effector and a media positioned within the jaws of the end effector based on the aforementioned sensing techniques. For example, according to the non-limiting aspect of FIG. **44**, an initial tissue contact can be detected **4402** via the aforementioned RF impedance sensing techniques. It shall be appreciated that the detection **4402** step of the method **4400** of FIG. **44** can be performed via any of the aforementioned methods, including those discussed in reference to FIGS. **6** and **13-15**.

[0202] According to the non-limiting aspect of FIG. **44**, the method **4400** can further include initiating **4404** a sensing-based characterization process based on the detection **4402** of the particular contact between the jaws of an end effector and a media positioned within the jaws of the end effector. For example, according to the non-limiting aspect of FIG. **44**, a tissue-thickness sensing method can be initiated **4404** based on the detected **4402** initial contact between the jaws of the end effector and the tissue. Of course, any number of sensors can be configured to characterize any number of media positioned within the jaws of the end effector in any way and thus, according to other non-limiting aspects, the initiated **4404** characterization can include any other sensing functions of any other sensors.

[0203] In further reference to FIG. **44**, the method **4400** can further include determining **4406** a tissue characteristic based on sensing method. For example, according to the non-limiting aspect of FIG. **44**, the method **4400** can further include determining **4406** a tissue thickness based on thickness sensing method. If the characteristic cannot be determined, the method **4400** can further include continually taking **4408** measurements and comparing the results to an initial baseline. However, if the characteristic can be determined, the method **4400** can further include outputting **4410** the determined characteristic. For example, according to some non-limiting aspects, the

determined characteristic can be output to a communicably coupled computing device, such as a surgical hub connected to the surgical instrument. However, according to other non-limiting aspects, the determined characteristic can be output to an on-board display of the surgical instrument, a cloud-computing device, and/or a mobile computing device of the technician, amongst other communicably coupled computing devices.

[0204] Likewise, information gained from the aforementioned sensing techniques performed by a first surgical instrument can be alternately applied to a second surgical instrument, illustrating the possibilities of a connected operating room environment. In other words, the present disclosure contemplates non-limiting aspects wherein information from previous firings of different devices could then be used to inform the firing of a second device, regardless of whether the second device is configured to perform the aforementioned sensing techniques. Whereas a fully “smart instrument ecosystem” may require sensors and communication modules installed within each instrument, such systems could result in inefficiencies and might drive costs beyond a point of acceptability where multiple surgical instruments are used in single procedure. Accordingly, it would be extremely beneficial if the aforementioned sensing techniques could be performed by a first surgical instrument and alternately applied to a second surgical instrument.

[0205] For example, information gained from the sensing capabilities of a first surgical instrument can be leveraged to inform the firing of one or more separate surgical instruments used during a procedure. This can be accomplished via algorithmic control of the firing sequence of a second device and/or the display of information to the surgeon. As such, the aforementioned sensing techniques can enable an operating clinician to benefit from their use of an entire range of surgical instrument, so long as at least one instrument is configured to generate insights, as previously disclosed. When combined with knowledge of the steps in a surgical procedure, on-screen prompts via a display system, including a heads-up display, could enable the sensed parameters from the first device (e.g., a smart instrument) to inform on-screen recommendations based on the anticipated next step and additional surgical instruments to be used in the procedure, regardless of whether or not those additional instruments are “smart.”

[0206] Referring now to FIG. 45, a logic flow diagram of a method 4500 of utilized the sensing techniques disclosed herein to inform an operation performed by a second surgical instrument is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. 45, the method 4500 can include sensing 4502, via a first surgical instrument, a tissue parameter via the aforementioned sensing techniques. For example, it shall be appreciated that the sensing 4502 step of the method 4500 of FIG. 45 can be performed via any of the aforementioned methods, including those discussed in reference to FIGS. 6 and 13-15. The method 4500 can further include communicating 4504 the sensed parameter to a computing device, such as a central hub, communicably coupled to the first surgical instrument.

[0207] Having communicated 4504 the sensed parameter, the method 4500 of FIG. 45 can further include determining 4506 if a second surgical instrument is required by the surgical procedure and whether or not the second surgical instrument is a “smart” instrument, meaning the second instrument is connected and able to receive inputs to inform its operation. Assuming the second surgical instrument is not a “smart” instrument, the method 4500 can further include anticipating 4510 a use of the second instrument based on procedure steps and informing 4512 an operation of the second instrument based on the parameter sensed by the first surgical instrument, and terminating 4518 the process. In other words, the hub itself can provide useful information for the operation of the second surgical instrument that are derived from the parameter sensed by the first surgical instrument via a display, such as a heads-up display, for example. Even though the insights generated by the first surgical instrument cannot be directly implemented via the second surgical instrument, they can be presented to the operating clinician to inform the operation.

[0208] However, assuming it is determined that the second surgical instrument is a “smart” instrument, the method 4500 of FIG. 45 can include transmitting 4514 the parameter sensed by the

first instrument to the second instrument and performing **4516** the aforementioned sensing techniques via the second instrument. The parameter sensed by the first instrument can be used to supplement or otherwise autonomously influence **4516** operation of the second surgical instrument. For example, the transmission **4514** can include some or all of the insights generated by the first surgical instrument and can inform firing, sensing, or other device functions performed by the second surgical instrument. In other words, because the second surgical instrument is connected, the second surgical instrument can autonomously influence its own operation based on algorithmic inputs provided by the first surgical instrument, without intervention from the operating clinician. Once complete, the method **4500** can include terminating **4518** the process. In other words, the method **4500** of FIG. **45** can be especially valuable if the sensed parameters from the first instrument can be used as inputs to algorithms used for sensing on the second instrument, because the method **4500** can enable the second instrument to have relevant data that it may not have been able to otherwise collect. In other words, only the first surgical instrument need to be configured to sense and generate insights, as previously described, but other devices—including both connected and disconnected device, alike—can benefit from those insights. Thus, a “smart instrument ecosystem” can be both effectively and efficiently achieved via the devices, systems, and methods disclosed herein.

[0209] Referring now to FIG. **46**, a logic flow diagram of a specific procedure **4600** implementing the method **4500** of FIG. **45** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **4600**, the procedure **4600** can include a colectomy, wherein the first surgical instrument includes an endocutter, for example, and the second surgical instrument includes a circular stapler, for example. Of course, FIG. **46** is merely illustrative of the method **4500** of FIG. **45** and, according to other non-limiting aspects, the procedure can include any number of surgical operations performed by any number of surgical instruments.

[0210] Nonetheless, the procedure **4600** of FIG. **46** can include sensing **4602** a tissue thickness and disease state via the aforementioned sensing techniques performed by the endocutter. For example, the sensing **4602** step can be performed via any of the aforementioned methods, including those discussed in reference to FIGS. **6** and **13-15**. The procedure can further include communicating **4604** the sensed tissue thickness and disease state to a communicably coupled computing device, such as a central hub. Having communicated **4604** the sensed parameter, the procedure **4600** can further include determining **4606** that the circular stapler is required by the colectomy and whether or not the circular stapler is a “smart” instrument. Assuming the circular stapler is not a “smart” instrument, the procedure **4600** can further include anticipating **4610** a use of the circular stapler based on procedure **4600** steps and informing **4612** an operation of the circular stapler based on the parameter sensed by the first surgical instrument and terminating **4618** the process. Once again, informing **4612** the operation of the circular stapler can include displaying insights generated by the endocutter to the operating clinician to inform the operation.

[0211] However, assuming it is determined that the circular stapler is a “smart” instrument, the procedure **4600** of FIG. **46** can include transmitting **4614** the tissue thickness and disease state to the circular stapler for implementation, which can include slowing **4616** the rate of fire and/or limiting a force to fire based on the thickness and disease state. For example, the circular stapler can employ algorithmic intelligence to determine that less speed and/or force to fire may be necessary to cut the diseased tissue. In other words, because the circular stapler is connected, the circular stapler can influence its operation without intervention from the operating clinician. Once complete, the method **4500** can include terminating **4518** the process. In other words, the method **4500** of FIG. **45** can be especially valuable if the sensed parameters from the first instrument can be used as inputs to algorithms used for operating the second instrument, because the method **4500** can enable the second instrument to have relevant data that it may not have been otherwise able to collect.

[0212] In other words, in the non-limiting example wherein the procedure 4600 is a colectomy, an endocutter may be used to cut out a portion of the colon, and then a circular stapler could be used for end to end anastomosis. If the circular stapler is a “dumb” device, but the endocutter is a “smart” device, a communicably coupled computing device, such as a central surgical hub, could interpret the insights generated by the endocutter in conjunction with inferred surgical procedure steps to make recommendations for something such as a staple cartridge selection for the circular stapler. If the circular stapler is a “smart” device, it could utilize the insights generated by the endocutter as algorithmic inputs to autonomously influence and/or improve operation of the circular stapler, as previously described.

[0213] While the foregoing methods, algorithms, instruments, and systems provided various examples of differentiating media grasped by a surgical end effector mainly based on impedance data, it is understood that the methods, algorithms, instruments, and systems of the present disclosure can be equally applied to other sensed parameters such as current, temperature, and pressure, for example. In at least one instance, for example, methods, algorithms, instruments, and systems can be utilized to differentiate tissue from a foreign object based on detected differences between tissue and a foreign object.

[0214] While several forms have been illustrated and described, it is not the intention of Applicant to restrict or limit the scope of the appended claims to such detail. Numerous modifications, variations, changes, substitutions, combinations, and equivalents to those forms may be implemented and will occur to those skilled in the art without departing from the scope of the present disclosure. Moreover, the structure of each element associated with the described forms can be alternatively described as a means for providing the function performed by the element. Also, where materials are disclosed for certain components, other materials may be used. It is therefore to be understood that the foregoing description and the appended claims are intended to cover all such modifications, combinations, and variations as falling within the scope of the disclosed forms. The appended claims are intended to cover all such modifications, variations, changes, substitutions, modifications, and equivalents.

[0215] The foregoing detailed description has set forth various forms of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, and/or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. Those skilled in the art will recognize that some aspects of the forms disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as one or more program products in a variety of forms, and that an illustrative form of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution.

[0216] Instructions used to program logic to perform various disclosed aspects can be stored within a memory in the system, such as dynamic random access memory (DRAM), cache, flash memory, or other storage. Furthermore, the instructions can be distributed via a network or by way of other computer readable media. Thus a machine-readable medium may include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer), but is not limited to, floppy diskettes, optical disks, compact disc, read-only memory (CD-ROMs), and

magneto-optical disks, read-only memory (ROMs), random access memory (RAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic or optical cards, flash memory, or a tangible, machine-readable storage used in the transmission of information over the Internet via electrical, optical, acoustical or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.). Accordingly, the non-transitory computer-readable medium includes any type of tangible machine-readable medium suitable for storing or transmitting electronic instructions or information in a form readable by a machine (e.g., a computer).

[0217] As used in any aspect herein, the term “control circuit” may refer to, for example, hardwired circuitry, programmable circuitry (e.g., a computer processor including one or more individual instruction processing cores, processing unit, processor, microcontroller, microcontroller unit, controller, digital signal processor (DSP), programmable logic device (PLD), programmable logic array (PLA), or field programmable gate array (FPGA)), state machine circuitry, firmware that stores instructions executed by programmable circuitry, and any combination thereof. The control circuit may, collectively or individually, be embodied as circuitry that forms part of a larger system, for example, an integrated circuit (IC), an application-specific integrated circuit (ASIC), a system on-chip (SoC), desktop computers, laptop computers, tablet computers, servers, smart phones, etc. Accordingly, as used herein “control circuit” includes, but is not limited to, electrical circuitry having at least one discrete electrical circuit, electrical circuitry having at least one integrated circuit, electrical circuitry having at least one application specific integrated circuit, electrical circuitry forming a general purpose computing device configured by a computer program (e.g., a general purpose computer configured by a computer program which at least partially carries out processes and/or devices described herein, or a microprocessor configured by a computer program which at least partially carries out processes and/or devices described herein), electrical circuitry forming a memory device (e.g., forms of random access memory), and/or electrical circuitry forming a communications device (e.g., a modem, communications switch, or optical-electrical equipment). Those having skill in the art will recognize that the subject matter described herein may be implemented in an analog or digital fashion or some combination thereof.

[0218] As used in any aspect herein, the term “logic” may refer to an app, software, firmware and/or circuitry configured to perform any of the aforementioned operations. Software may be embodied as a software package, code, instructions, instruction sets and/or data recorded on non-transitory computer readable storage medium. Firmware may be embodied as code, instructions or instruction sets and/or data that are hard-coded (e.g., nonvolatile) in memory devices.

[0219] As used in any aspect herein, the terms “component,” “system,” “module” and the like can refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution.

[0220] As used in any aspect herein, an “algorithm” refers to a self-consistent sequence of steps leading to a desired result, where a “step” refers to a manipulation of physical quantities and/or logic states which may, though need not necessarily, take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It is common usage to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like. These and similar terms may be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities and/or states.

[0221] A network may include a packet switched network. The communication devices may be capable of communicating with each other using a selected packet switched network communications protocol. One example communications protocol may include an Ethernet communications protocol which may be capable permitting communication using a Transmission Control Protocol/Internet Protocol (TCP/IP). The Ethernet protocol may comply or be compatible with the Ethernet standard published by the Institute of Electrical and Electronics Engineers (IEEE) titled “IEEE 802.3 Standard”, published in December 2008 and/or later versions of this standard.

Alternatively or additionally, the communication devices may be capable of communicating with each other using an X.25 communications protocol. The X.25 communications protocol may comply or be compatible with a standard promulgated by the International Telecommunication Union-Telecommunication Standardization Sector (ITU-T). Alternatively or additionally, the communication devices may be capable of communicating with each other using a frame relay communications protocol. The frame relay communications protocol may comply or be compatible with a standard promulgated by Consultative Committee for International Telegraph and Telephone (CCITT) and/or the American National Standards Institute (ANSI). Alternatively or additionally, the transceivers may be capable of communicating with each other using an Asynchronous Transfer Mode (ATM) communications protocol. The ATM communications protocol may comply or be compatible with an ATM standard published by the ATM Forum titled "ATM-MPLS Network Interworking 2.0" published August 2001, and/or later versions of this standard. Of course, different and/or after-developed connection-oriented network communication protocols are equally contemplated herein.

[0222] Unless specifically stated otherwise as apparent from the foregoing disclosure, it is appreciated that, throughout the foregoing disclosure, discussions using terms such as "processing," "computing," "calculating," "determining," "displaying," or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

[0223] One or more components may be referred to herein as "configured to," "configurable to," "operable/operative to," "adapted/adaptable," "able to," "conformable/conformed to," etc. Those skilled in the art will recognize that "configured to" can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

[0224] The terms "proximal" and "distal" are used herein with reference to a clinician manipulating the handle portion of the surgical instrument. The term "proximal" refers to the portion closest to the clinician and the term "distal" refers to the portion located away from the clinician. It will be further appreciated that, for convenience and clarity, spatial terms such as "vertical", "horizontal", "up", and "down" may be used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

[0225] Those skilled in the art will recognize that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as "open" terms (e.g., the term "including" should be interpreted as "including but not limited to," the term "having" should be interpreted as "having at least," the term "includes" should be interpreted as "includes but is not limited to," etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases "at least one" and "one or more" to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles "a" or "an" limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases "one or more" or "at least one" and indefinite articles such as "a" or "an" (e.g., "a" and/or "an" should typically be interpreted to mean "at least one" or "one or more"); the same holds true for the use of definite articles used to introduce claim recitations.

[0226] In addition, even if a specific number of an introduced claim recitation is explicitly recited,

those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

[0227] With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flow diagrams are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

[0228] It is worthy to note that any reference to “one aspect,” “an aspect,” “an exemplification,” “one exemplification,” and the like means that a particular feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. Thus, appearances of the phrases “in one aspect,” “in an aspect,” “in an exemplification,” and “in one exemplification” in various places throughout the specification are not necessarily all referring to the same aspect. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner in one or more aspects.

[0229] Any patent application, patent, non-patent publication, or other disclosure material referred to in this specification and/or listed in any Application Data Sheet is incorporated by reference herein, to the extent that the incorporated materials is not inconsistent herewith. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

[0230] In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more forms has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more forms were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various forms and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

Claims

1. A surgical instrument, comprising: an end effector comprising jaws configured to transition between an opened condition and a closed condition; a plurality of electrodes positioned within the jaws of the end effector, wherein each electrode of the plurality of electrodes is positioned about a longitudinal axis defined by the end effector; a control circuit and a memory configured to store an algorithm configured to cause the control circuit to: receive signals from the plurality of electrodes; determine an impedance signal based on the signals received from the plurality of electrodes; detect a media positioned between the jaws of the end effector based on the determined impedance signal; determine a position of the detected media along the longitudinal axis based on the signals received from the plurality of electrodes; and generate an alert associated with the detected media and the determined position.
2. The surgical instrument of claim 1, wherein the alert comprises a digital representation of the end effector, the detected media, and the determined position, and wherein the algorithm is further configured to cause the control circuit to initiate a transmission of the digital representation to a display for visual review by the operating clinician.
3. The surgical instrument of claim 1, wherein the alert comprises an audible alert, and wherein the algorithm is further configured to cause the control circuit to cause a speaker communicably coupled to the control circuit for audible transmission to the operating clinician.
4. The surgical instrument of claim 1, wherein the algorithm is further configured to cause the control circuit to: determine a second impedance signal based on the signals received from the plurality of electrodes; detect a second media positioned between the jaws of the end effector based on the determined second impedance signal; and determine a second position of the detected second media along the longitudinal axis based on the signals received from the plurality of electrodes; and wherein the alert is also associated with the detected second media and the determined second position.
5. The surgical instrument of claim 4, wherein the algorithm comprises a linear support vector machine configured to distinguish the media from the second media based on the determined impedance signal and the determined second impedance signal.
6. The surgical instrument of claim 4, wherein the algorithm detected media comprises a tissue sample, and wherein the detected second media comprises a foreign object.
7. The surgical instrument of claim 1, wherein the algorithm is further configured to cause the control circuit to characterize the detected media based on the determined impedance signal.
8. The surgical instrument of claim 1, wherein the impedance signal comprises an impedance magnitude.
9. The surgical instrument of claim 1, wherein the impedance signal comprises an phase.
10. The surgical instrument of claim 1, wherein the algorithm is further configured to cause the control circuit to determine a change in the impedance signal across the time domain.
11. The surgical instrument of claim 1, wherein the algorithm is further configured to cause the control circuit to determine a change in the impedance signal across the frequency domain.
12. A surgical system, comprising: a surgical instrument comprising: an end effector comprising jaws configured to transition between an opened condition and a closed condition; and a plurality of electrodes positioned within the jaws of the end effector, wherein each electrode of the plurality of electrodes is positioned about a longitudinal axis defined by the end effector; and a computer system communicably coupled to the surgical instrument, wherein the computer system comprises a control circuit and a memory configured to store an algorithm configured to cause the control circuit to: receive signals from the plurality of electrodes; determine an impedance signal based on the signals received from the plurality of electrodes; detect a media positioned between the jaws of the end effector based on the determined impedance signal; determine a position of the detected

media along the longitudinal axis based on the signals received from the plurality of electrodes; and generate an alert associated with the detected media and the determined position.

13. The surgical system of claim 12, wherein the algorithm is further configured to cause the control circuit to: determine a second impedance signal based on the signals received from the plurality of electrodes; detect a second media positioned between the jaws of the end effector based on the determined second impedance signal; and determine a second position of the detected second media along the longitudinal axis based on the signals received from the plurality of electrodes; and wherein the alert is also associated with the detected second media and the determined second position.

14. The surgical system of claim 13, wherein the algorithm detected media comprises a tissue sample, and wherein the detected second media comprises a foreign object.

15. The surgical system of claim 12, wherein the algorithm is further configured to cause the control circuit to characterize the detected media based on the determined impedance signal.

16. A method of characterizing media positioned between jaws of an end effector of a surgical instrument, the method comprising: receiving, via a control circuit of the surgical instrument, signals from a plurality of electrodes positioned within the jaws of the end effector; determining, via the control circuit, an impedance signal based on the signals received from the plurality of electrodes; detecting, via the control circuit, the media positioned between the jaws of the end effector based on the determined impedance signal; determining, via the control circuit, a position of the detected media along a longitudinal axis defined by the end effector based on the signals received from the plurality of electrodes; generating, via the control circuit, an alert associated with the detected media and the determined position; and characterizing, via the control circuit, the detected media based on the determined impedance signal.

17. The method of claim 16, further comprising: determining, via the control circuit, a second impedance signal based on the signals received from the plurality of electrodes; detecting, via the control circuit, a second media positioned between the jaws of the end effector based on the determined second impedance signal; and determining, via the control circuit, a second position of the detected second media along the longitudinal axis based on the signals received from the plurality of electrodes; and wherein the alert is also associated with the detected second media and the determined second position.

18. The method of claim 17, further comprising distinguishing, via the control circuit, the media from the second media based on the determined impedance signal and the determined second impedance signal in accordance with a linear support vector machine.

19. The method of claim 16, wherein generating the alert comprises: generating, via the control circuit, a digital representation of the end effector, the detected media, and the determined position; and initiating, via the control circuit, a transmission of the digital representation to a display for visual review by an operating clinician.

20. The method of claim 16, wherein the alert comprises an audible alert, and wherein the method further comprises causing, via the control circuit, a speaker communicably coupled to the control circuit to play the audible alert.
